

Golf Placement Robot, using the AL5D Robotic Arm



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Contents

Contents	2
Abstract	4
Introduction	4
Objectives	5
Aims	6
Scope	6
Methodology.....	8
Project Planning and Timelines	8
Gantt Chart	8
Activity List.....	9
Work Breakdown Structure.....	10
Work Distribution	10
Project Scheduling and Monitoring.....	11
Critical Path	11
Stakeholder Analysis.....	12
Stakeholder Matrix.....	12
Stakeholders	12
Risk Analysis	14
Risk Considerations.....	14
Risk Management & Mitigation Plan	14
Physical Risk Assessment for Team Members.....	15
Physical Risk Assessment for Users	16
Financial Analysis	17
Funding.....	17
Financial Considerations	17
Financial Plan.....	17
Design Fundamentals	18
High-Level Design.....	18
Low-Level Design	18
Validation and Verification.....	19

System Verification.....	19
System Validation.....	19
Verification and Validation Plan	19
Verification Methods.....	19
Validation Methods.....	20
Gripper Design.....	22
Design 1.....	22
Design 2.....	23
Design 3.....	24
Design 4.....	25
Design 4 – Redesigned	27
Design 5.....	27
High-Level Design Discussion:	28
High-Level Gripper Design Analysis	30
Gripper Design Verification	31
Gripper Manufacturing	32
Gripper Manufacturing Considerations	32
Manufacturing Technique Verification.....	33
3D Printing Material Considerations	35
Material Verification	36
Robot CAD Model	36
Peripheral Design	38
Robotics Design	46
Hardware.....	46
Software	47
Requirements.....	48
Reflection	49
References.....	49

Abstract

This report follows the design and engineering process and steps taken in prototyping a robotic arm designed to take and place standard golf balls in a range for consecutive hits in a drive range. The robot used is a LYNX motion AL5D robotic arm. After a few group brainstorm sessions and a few design iterations a final grip design which is optimized for gripping golf balls was finalized and tested along with the code for automated function which was created using MATLAB. Using standardized design procedures, the steps have all been analysed in this report, including the scope of the project, the methodology used, safety precautions taken, stakeholder specifications, risk assessments and cost analysis of the project.

Introduction

Automation is a vital component in many fast-paced industries for overall efficiency and growth as it saves time and cost by executing tasks accurately and with high precision repeatedly. The golf industry has been slow in integrating automation as the game has been roadblocked due to the nature of the game, where the game is played in a large outdoor range making it impractical. However, in a driving range or in a mini golf course where the golf balls are consistently hit from a set location it becomes necessary to have a golf ball loading system for better customer experience and satisfaction. A robotic arm which has been programmed and precisely calibrated to place the golf balls on the target location can be easily implemented for a better experience.

The AL5D Robotic Arm delivers fast, accurate and repeatable movement, with a time-tested rock-solid design. It uses the Servo Erector Set components for ultimate flexibility and expandability. (Nantel, 2024). This makes it the perfect arm for developing a new innovative design, to aid with many different industries. For this project, 4 innovative designs were suggested, and within the report, one of these ideas will be discussed, a pick-and-place robot for a golf range. The design brief suggests that this robot must pick up a golf ball and place it onto a tee in front of the golfer, allowing faster, repeated swings without stopping. However, upon research, it was found that golf ranges do not use golf tees. This robot, therefore, will be capable of placing the golf ball on a specified, adjustable point, or on a tee to conform for all golfing scenarios. A robotic arm which automatically places the ball quickly will be most beneficial to professional golf players, who are aiming to advance their swing through repetition, or beginners who are trying to learn the sport.

Aims

Objectives

The main objectives of this report is in making a golf ball placing mechanism for a driving range, in order to achieve this the factors considered were convenience, practicality, repeatability, integration and safety.

Any user interface should be **Conveniently** accessible and understandable by the user, to ensure this the robot system should be simple with each part of any complex process being able to be executed with minimum user input. Also, the system should have clear feedback of the process taking place in real time.

When transferring from paper/virtual design to real world function of a system certain **Practical** aspects of it need to be considered such as, how fast should the ball be replaced, what type of grip would allow the maximum accuracy with picking up, moving and placing a standard golf ball, how to efficiently communicate to the user and how would the robot stay out of the way once its task has been completed.

Once a functional single unit is created, it should be able to be easily **Replicated/Repeated** in both hardware and software functionality for mass installation. Hence, standardized machine code and readily available robotic components would be ideal.

To **Integrate** this system in golf ranges it should be in accordance with the requirements of the stakeholders, who would be very keen on making sure that all bases of concern are covered for within the design and the model can be smoothly deployed, maintained and operated with minimum redesign to the golf range and in a convenient manner.

Many **Safety** precautions need to be taken into consideration as its paramount that no individual is put at risk while in operation. Due to that fact any all hazardous scenarios should be taken into account and countered or avoided.

Scope

The scope of the project focuses on developing a golf ball pick and place robot system designed to enhance operational performance, user experience and integration with existing golf club facilities. The primary aim is to automate the placement of golf balls on the tee smoothly in short time and to improve the operation system of the golf club and reliability with high safety standards.

The key deliverables of the project include detailed mechanical designs as CAD models, electrical components overview, software documentation as codes used to make robot movements, a product design specification with system requirements, testing reports and validation videos.

The project timeline includes the critical tasks like the design consideration of the golf ball transfers from storage to the robot arm and monitoring systems utilizing sensors and webcam for accuracy, safety, and user-friendliness. As an example, team members are investigating the effective way of transition of golf ball in short time without deviating from its path and development of customizable gripper and robot arm to operate appropriately with sensors integration. The robot arm is being programmed to pick and place golf balls efficiently in a precise, reliable manner. As safety is the top priority, sensors and webcams are used to monitor the robot's movements and potential failures. Iterative processes of testing and validation reinforce the project, ensuring the working of each function as intended and contributing to the overall project's success.

While the project is designed to carry out a high standard of automation, several limitations should be noted. For instance, the current system is for standard golf balls and may encounter challenges with other ball sizes. The system does not account for some extreme environmental conditions during outdoor use as heavy rain or heavy snow and irregular surfaces which can affect the robot's performance. Regular human assistance should be needed for kids and beginners. Future iterations may address these limitations, improving robot's safety, reliability, and sustainability.

Although these constraints, the project is designed to deliver a reliable, efficient, and user-friendly robot system. The robust approach to design, good implementation and iterative testing ensures the project's success in reaching the objectives and requirements.

Scope by Josh...

The primary objective of this project is to make a fully automatic and precise golf ball dispensing and placing system in a driving range in a safe and cost-effective ways to aid players with the hassle of constant bending for setting up shots, help in saving time and

to improve the overall quality of gameplay. Towards this each step of the design process was given considerable thought and scrutiny, and multiple iterations of the same elements was compared to settle on the best result.

With the initial progress being shown to the stakeholders when a working prototype with the some of the primary functions for repeated golf placing can be performed by the arm with no functional errors. This would require completed grip design which would be modelled and printed and a sample software of the basic movement on a controlled loop.

When the arm can pick up and place a ball precisely for a strike while avoiding being in the way of the strike and being able to repeat this on command would be considered the minimum threshold for a successful demonstration to the stakeholders before more resources are to be allocated for further development and final production.

To achieve this in given the timeline of the project the workload has been dispersed among individuals with consideration to their specialized academic fields. Where the tasks would be done by the individuals suited most for them. An easy way of presetting motion needed to be developed where the location for pickup up and placing the golf ball can be easily given by the user to the robot which then will save and repeat the motion as needed. For this 'Teach Mode' was conceptualized where within the given range of motion of the arm any sequence of motion can be easily recorded and run on command.

For factors of safety the robotic arm was given failsafe where the code would stop immediately on the push of a button and terminate all functions in case of unforeseen errors that might occur. Other safety measures include indications to the user when and what functions are being performed, for this many indications methods were considered from LED to audio frequencies, it was settled on an audio message being played of the exact function. For example, when the robot is about to move the software will output 'robot in motion', to have clear indication to the user so as to avoid any confusion.

The user interface is also kept simple and would consist of a guide for ease of use so anyone can operate the system within a few minutes of reading. Before this every function will be tested and validated to ensure smooth operation of the user interface and system function.

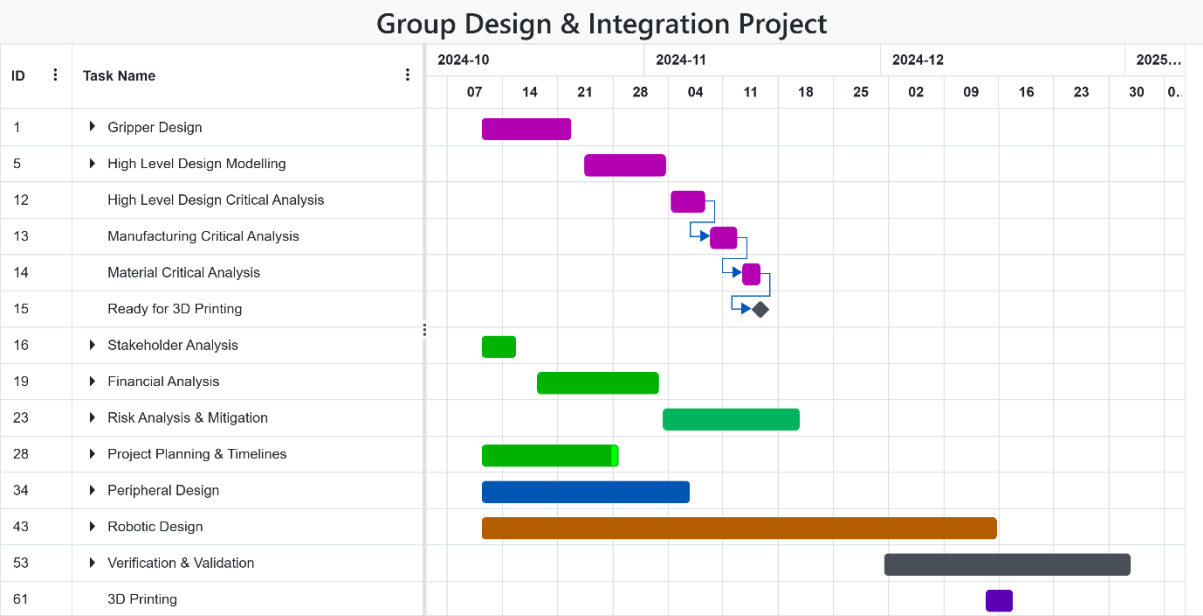
The robotic arm would be designed to work precisely in ideal conditions like flat ground and no wind but if these factors were to apply the ball could roll away on setting down. In this scenario the results would vary, and it's excluded in this study, but later iterations could potentially solve the problem. However, these would not affect in a typical driving range and the effected can be considered negligible in comparison to the rest.

Methodology

To complete the stated aim of this project the following methods were used... To code and program the robot, MATLAB was used, and the system was validated and worked smoothly by combining a CAD model of the arm into the system. Hand drawings were used to initially create ideas for the new gripper design, as well as the peripheral. This was then translated into CAD models using SolidWorks, before being critically analysed to ensure the best design was used for the scope. Critical analysis was also completed to assess which control board, power supply and electronic system was best. Both the gripper and peripheral design were functionally tested by using a to-scale golf ball CAD model, to ensure the dimensions were accurate. The gripper design was then 3d printed before being filed down to ensure no sharp edges were present to reduce the safety risk when attaching to the original gripper design. The peripheral (golf ball shelf) was manufactured using hand tools, such as a point file, saw and glue. This method was found to be best, due to the shape of the peripheral, and the complexity it would bring if 3d printed.

Project Planning and Timelines

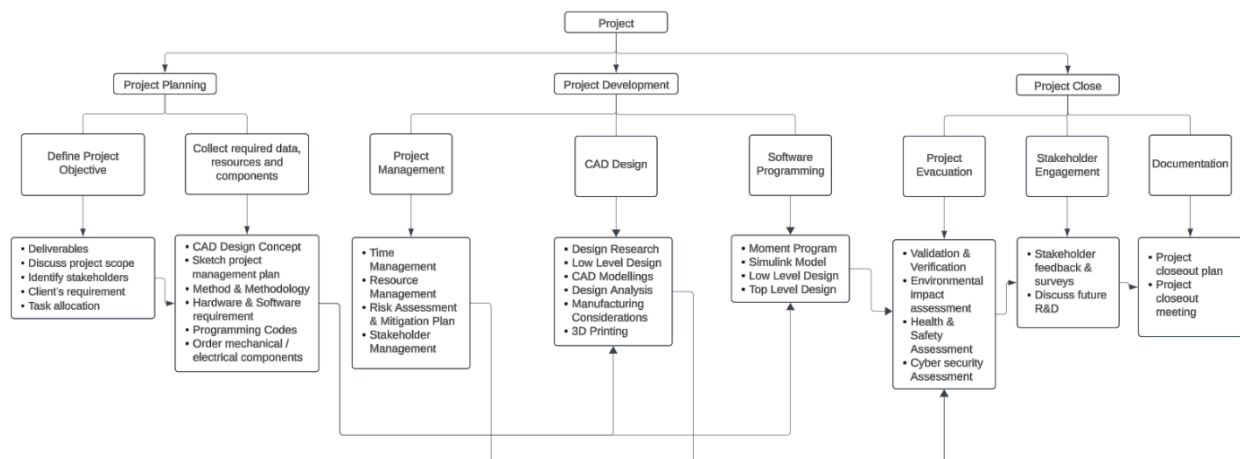
Gantt Chart



Activity List

Project Planning	Define project objectives	Deliverables Discuss project scope Identify stakeholders Client's requirement Sketch project management Task allocation
	Collect required data, resources, and components	CAD design concept Method & Methodology Hardware & Software requirement Programming codes
Project Development	Project Management	Time Management Resource Management Risk Assessment & Mitigation Plan Financial Plan Stakeholder Management
	CAD Design	Design Research Low Level Design CAD Modelling Design Iteration & Selection Manufacturing Considerations 3D Printing
	Software Programming	Movement program Simulink Model Low Level Design Top level design
Project Close	Project Evacuation	Validation and Verification Environmental impact Assessment Health & Safety Assessment Cyber Security Assessment
	Stakeholder Engagement	Stakeholder Feedback & Surveys Discuss future R&D for the product
	Documentation and Reporting	Project Closeout Plan Project Closeout Meeting

Work Breakdown Structure



Work Distribution

To extrude the maximum output for the project, tasks were divided into CAD Design and Modelling, Project Management, 3D printing and Software Programming based on the individual's skills and expertise. Mechanical students would take CAD Modelling, Project Management, 3D printing and extra tasks while the robotic student would lead software development. All students then performed verification and validation to lead to the project's success.

Tasks were first listed in the Activity List to start initial planning, then structured in the Work Breakdown Structure (WBS). The Critical Path is created to present the most important project plan, the work after modelling the gripper design and programming the robot operation before verification. Gantt Chart was used to structure the project timeline. The WBS is open to both project team and stakeholders, keeping everyone participated in all stages through the project.

Project plans are divided into project planning, project development and project close. In project planning, definition of project objectives as project scope, deliverables and stakeholder's requirements, and collection of required data and resources such as CAD design concept, method and methodology, project management plan and order of components.

Project development is a group of certain tasks. According to WBS, exploration of project management is timeline, funding and finance, stakeholder management, risk assessment and mitigation plan. Looking at CAD design, simultaneous tasks as design research, implementation of multiple designs, critical analysis of these designs and manufacturing consideration of final design.

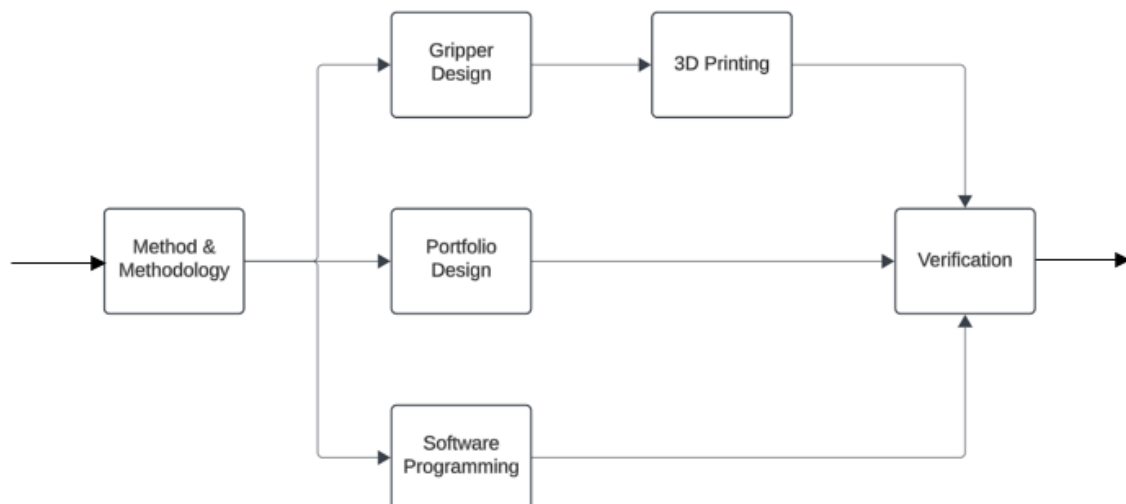
Project close is designed to execute verification and validation, environmental impact, health and safety, and cyber security assessment. Stakeholder engagement needs to be conducted to receive feedback and surveys from users and golf clubs

and discuss future research and development plans. Project documentation is finally done when the robot is handed over to customers.

Project Scheduling and Monitoring

To make sure we catch up with the milestones of the project, we consistently monitored team members' progress using Meeting Notes and Gantt Chart. These allowed us to efficiently plan and schedule the work we had completed and were going to do before the next meeting. We had a team meeting every week. During these meetings, each team member provided updates on their tasks such as CAD modelling, coding and project management. We discussed the research, issues and solutions on specific tasks. Then we conducted the tasks we were going to do for the coming week.

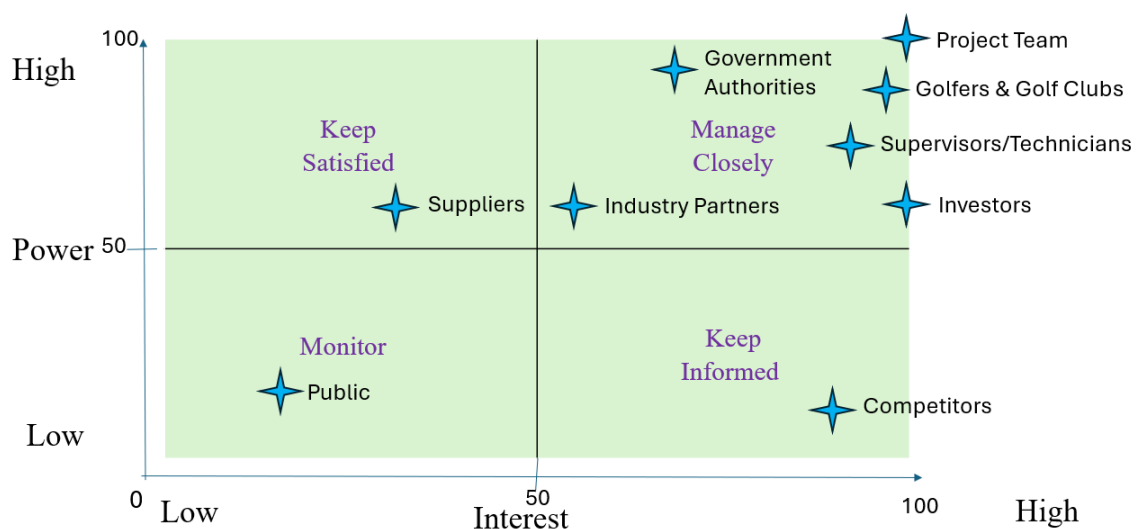
Critical Path



Stakeholder Analysis

Stakeholder engagement was conducted at the beginning of the project meetings by identifying and organizing all relevant sectors, internal and external stakeholders. A power-interest matrix is used to describe based on their influence and attention in the project. With this matrix, the project team can decide which stakeholders were important and engage strategies to communicate regularly while appropriately approaching other groups.

Stakeholder Matrix



Stakeholders

Project Team – The project team is responsible for CAD design, hardware, software programming, verification, and validation throughout the whole project to capture project success.

Project Supervisors and Technicians – Supervisors provide guidelines and ensure adherence to engineering standards while technicians assist with 3D printing and testing. For example, – Module Team from UWE, Technicians from 3D printing.

Golfers and Golf Clubs – Golf players and clubs are clients and primary users, providing feedback to ensure the robot functions meet the requirements. For example, The Bristol Golf Club, Henbury Golf Club, Kendleshire Golf, etc.

Investors – Shareholders provide funding for the project development and future R&D sectors. For example, Innovate UK, Angel Investors and golf equipment manufacturers like Titleist, Callaway, etc.

Government Authorities – Regulatory Organizations ensure compliance with health, safety, environmental and data protection regulations. For example, Health and Safety Executive (HSE), National Cyber Security Center (NCSC), Bristol City Council, etc.

Commercial Clients – Clients purchase robot systems and help tailor to market demands. For example, Topgolf, The Bristol Golf Club, etc.

Industrial Partners – Partners offer technical support and integration expertise to project completion and future development. For example, robot systems manufacturers as Fanuc or KUKA Robotics and golf tech Companies as TrackMan, Flightscope, etc.

Suppliers – Suppliers ensure access to hardware needs, mechanical components, and materials for project. For example, Farnell, RS Components, Ultimaker, Amazon.

Competitors – Competitors highlight industrial standards and areas for improvements. For example, Power Tee (automated golf tee systems), Golf Tech Startups, etc.

Public – The public provides insights and feedback to make the system more sustainable and user-friendly. For example, amateurs, families and friends, etc.

Risk Analysis

Risk Considerations

To ensure risk awareness in our project, we evaluated and determined the potential risks in every sector. It involved a preliminary risk analysis overall the whole project as, technical {design, machinery parts, software, and sensors), time delay, financial, sustainability (environmental impact, health, and safety), and stakeholder consistency and communication. After identifying the hazards, we assessed the risks into severity and likelihood of each risk, and we planned how to mitigate the risks effectively. We also examined physical risk to both the team members and users how to perform additional control measures to take out the risks.

Risk Management & Mitigation Plan

Risk	Impact	Likelihood	Severity	Rank	Mitigation Plan	Likelihood After Mitigation	Severity After Mitigation	Rank After Mitigation
Machinery Breakdown	The robot may stop operating or cause project delays.	3	5	15	Regular check on mechanical components and make spare parts available in short period.	2	4	8
Software Bugs	Incorrect behaviour and delay in robot operation.	2	5	10	Start the coding early to get time to verify. Review the codes regularly while in process.	2	3	6
Sensor Failure	Inaccurate detection of golf ball and wrong golf ball placements.	2	5	10	Test various sensor types and select the most reliable option.	2	2	4
Limited Robot Arm Length	Incomplete functionality.	2	4	8	Consider adjusting tee placement to modify when needed	2	2	4
Budget Overrun	Unable to buy required components.	2	3	6	Record the spending closely, explore the cheaper alternative and prioritize the essential components.	1	2	2
Improper Time Management	Miss deadlines or can impact on project's	3	4	12	Make a detailed project timeline,	2	3	6

	quality because of rushing.				report weekly report to team members.			
Poor Resource Management	Inefficient use of 3D printer slots can cause delays and extra costs.	2	3	6	Allocate team members to specific resources, learn or contribute between team members to use the resources efficiently	1	2	2
CAD Design Problem	Can lead to project failure.	2	5	10	Produce the first design early, test frequently and make improvements.	2	2	4
Inconsistency with Stakeholders	Misalignment with client's expectations could cause in scope changes and project delays.	2	3	6	Make regular meetings with clients and receive ongoing feedback.	1	2	2

Physical Risk Assessment for Team Members

Hazard Identified	Existing Control Measures	S	L	Risk Level	Additional Control Measures	S	L	Risk Level
Injury from parts, e.g. 3D printing, testing prototype robot	Wear PPE like gloves, goggles, and safety boots.	4	3	12	Share safety precaution reports to team members	2	2	4
Electric shock from components when testing	Check electrical components regularly and wear PPE when testing.	4	2	8	Share electrocution reports to team members	2	1	2
Eye problem from using computers for long time	Take a rest or nap after using the screen for 1 hour.	3	3	9	Activate eye protection features on devices	2	1	2
Slips or trips or Injury due to sharp debris on the floor from tools, broken parts.	Wear safety boots and regularly clean up the workspace and floor area.	3	4	12	Set the separate waste bin for these kinds of debris.	1	2	2

Physical Risk Assessment for Users

Hazard Identified	Existing Control Measures	S	L	Risk Level	Additional Control Measures	S	L	Risk Level
Accidents from moving robot parts	Define robot operating area and provide user instructions	5	3	15	Place the physical barriers to separate and access emergency stop button.	2	2	4
Electric shocks due to environmental conditions	Regular inspection of components and make sure robot operational area dry.	4	3	12	Cover the components with shock resistant materials. Add signs around high-risk components.	2	2	4

Risk Matrix

Very likely 5	5	10	15	20	25
Likely 4	4	8	12	16	20
Possible 3	3	6	9	12	15
Unlikely 2	2	4	6	8	10
Extremely unlikely 1	1	2	3	4	5
Likelihood (L) ↑ Severity (S) →	Minor injury – No first aid treatment required 1	Minor injury – Requires First Aid Treatment 2	Injury - requires GP treatment or Hospital attendance 3	Major Injury 4	Fatality 5

Action Level

POINTS:	RISK LEVEL:	ACTION:
1 – 2	NEGLIGIBLE	No further action is necessary.
3 – 5	TOLERABLE	Where possible, reduce the risk further
6 – 12	MODERATE	Additional control measures are required
15 – 16	HIGH	Immediate action is necessary
20 – 25	INTOLERABLE	Stop the activity/ do not start the activity

Financial Analysis

Funding

When investigating the budget of the project, multiple fundings have been explored, including external investors and possible grants. Initially, partnership with local golf clubs is an opportunity to secure partial funding through prototype trials, early access and discounts on purchases. Collaboration with golf technology companies is another option to get some budget and technical support at the beginning of the project. Additionally, some grant programs such as Innovate UK, British Robotics Seed Fund and EU Horizon Europe Grants are considered for significant funding to cover R&D sector, testing and transition to 'made-to-stock' These approaches ensure both financial sustainability and strong stakeholder engagement throughout the project.

Financial Considerations

To ensure the robot system is profitable and competitive, a detailed financial analysis is conducted. This consideration involved comparing the fixed costs and variable costs as opposed to potential -profitability over the project timeline.

The total fixed costs include labor, utilities, development and marketing, amount to £5,850, while variable costs per unit, such as material and components are estimated at £500. The selling price of £800 per unit makes a gross profit margin of 37.5% providing affordability and profitability for the golf clubs. With an initial investment of £200,000, the project is planned to break even after the sale of 20 units. A 2-year timeline is expected, and ROI stands at 72.375% after 2 years, showing strong financial potential. The robot will be manufactured using a 'made-to-order' model to minimize the budget needs and adapt to stakeholders' requirements.

Financial Plan

Fixed Cost	Cost (£)	Variable Cost	Cost (£)	Fixed Costs per Day (£)	Variable Costs per Unit	Selling Price per Unit (£)	Units Produced per Day	Break-Even Analysis
Labor Cost	3000	Robotic Arm Components	300	5850	500	800	1	20
Tools & Equipments	300	Sensors	50					
Workspace & Utility Costs	1000	Maintenance Costs	30	Initial Investment	Annual Profit	Return On Investment (ROI)		
Research & Development Cost	800	Delivery	50	200000	144750	72.375		
Safety Equipment Cost	200	3D Printing Materials	20					
Software Licensing	250	Additional Costs	50					
Marketing Costs	300	Total	500					
Total	5850							
				Item	Year 1	Year 2	After 2 Years	
				Fixed Costs	5850	0	5850	
				Variable Costs	125500	125500	251000	
				Total Costs	131350	125500	256850	
				Selling Price	200800	200800	401600	
				Annual Profit	69450	75300	144750	

Design Fundamentals

High-Level Design

Our high-level design planning was done with the plan to define the necessary steps, work division to ensure appropriate collaboration and creating a sample design diagram.

The first step was to give everyone tasks in the high-level design and filling in the meeting minutes as we went on. For example, we recorded the amount of work each person does to the meeting minus and what is expected of them by the next week.

- Meeting Minutes in Supporting Documents.

Low-Level Design

During the low-level design phase of the project, all requirements were addressed, and the successful integration of software and hardware components was ensured. Hardware and software were considered separately within the low-level design, which allowed for the confirmation of their successful integration. The specification included detailed descriptions of how each component should function, ensuring alignment with requirements at every stage. To guarantee that the low-level requirements were met, the requirements document was strictly followed.

A holistic approach was applied to achieve a robust, elegant, and well-documented solution. The process began with a thorough analysis of the project requirements, addressing both functional and non-functional aspects such as scalability and performance. Parallel design was prioritized, enabling all team members to work simultaneously on various tasks. This approach minimized workflow disruptions and improved the efficiency of the development process.

System co-design played a critical role, and significant research was conducted to optimize the final system. The development process involved discussions and revisions of system functionality, modelling of external factors, aligning system models with these factors, and analysing their potential impact on the system.

During co-simulation, smaller portions of hardware and software were simulated individually. Hardware simulations verified proper functionality, while joint simulations of hardware and software in MATLAB demonstrated the overall working principle of the robot. These simulations showcased how the system would operate in real-world scenarios, including the potential for accidents. Adjustments, such as altering paths, were tested to minimize resource waste and prevent injury.

Validation and Verification

System Verification

Verification addressed the complexity of the system using methods such as reachability, time, behaviour, and repeatability analyses. Reachability analysis determined whether certain states could be achieved from the system's initial state. This process started with a simple diagram of a driving range featuring a golf ball, robot, arm, and person, followed by MATLAB simulations.

Time analysis assessed the system's behaviour over time, which included proper path planning to ensure efficient use of time. Paths were adjusted and optimized for minimal travel distance. Behavioural analysis identified and evaluated any unusual or undesired behaviours, and these insights were used to refine the model, ensuring alignment with specifications and requirements. Repeatability analysis verified that the process could be performed multiple times without significant difficulty. This was particularly applied to the machine learning model for vision, ensuring it could consistently differentiate between areas with and without golf balls.

System Validation

Unit testing was essential throughout the project, with tests conducted at every stage of development. The initial robot was tested for functionality, followed by tests involving the integration of the camera and machine learning model, and further tests after full integration. These tests included basic control validation in MATLAB, manual calculations of forward kinematics, and extensive integration tests to reduce errors and improve repeatability. This approach progressed from simulations to a fully functional prototype.

Verification and Validation Plan

Verification:

- Requirement verification
- Design verification
- Code review
- Unit testing

Validation:

- Performance testing
- Functionality testing
- User acceptance testing

Verification Methods

Requirement Verification:

This method involves comparing the system and technical requirements to the stakeholder requirements and comparing the stakeholder requirements to the stakeholder interviews and research. This method ensures that all requirements are correct and meet the stakeholder standards. This method also ensures that the product meets industry-set standards, such as corrosion resistance or health and safety standards. This method was completed by having all group members assess the requirements to ensure they are consistent with the stakeholder interview data, as well as ensuring that the requirements set are concise.

Design Verification:

'During design verification, you verify that [the] design outputs meet [the] design inputs' (Lyons, 2023). This method compares all the product requirements to the design. This can be done using the below format and ensures that each design has a score which justifies the number of requirements met. This then shows which design is the best for the use, with possible improvements if there are no designs which meet the requirements enough.

Code Review:

This method 'acts as a quality assurance of the code base' (GitLab, 2023c). This ensures that the code is of a high standard; by having it reviewed by multiple peers, to ensure there have not been any errors missed by the initial personnel. This review aids in identifying bugs, logic problems and any other issues which may be present, before running the robot arm with the code. This allows for continuous improvements of the code while developing the robot, as well as ensuring there are no issues when the robot is being tested, and when a trial run is conducted.

Unit Testing:

'A unit test is a way of testing the smallest piece of code that can be logically isolated in a software application' (Smartbear, 2022). Testing small sections of code allows a granular view of how it is performing. Doing this ensures that all sections of code function to its most effective, as well as indicates where in the code there may be errors during runs, or if there is an issue with how the code was written. This allows the developer to check the work, before it is complete, ensuring that all sections of code are correct, as well as beneficial to the overall product.

Validation Methods

User Acceptance Test:

User acceptance testing can also be called Beta testing, or end-user testing and is a 'phase of software development in which the software is tested in the "real world" by the intended audience or business representative' (Stanford University, 2024). This test is aimed to ensure that the product meets the needs, requirements and wants of the consumer, before being released into the market and production. It is ideal to use this method when the final prototype is created as well as the final product. For this

project, a product user (golfer) will be using this product in an open space to simulate the “real world” situation.

Performance Test:

Performance testing is the practice of evaluating how a system performs in terms of responsiveness and stability under a particular workload (narwhal, 2018). These tests are normally performed to look at robustness, application size and reliability. The test includes indicators like, servo rest processing times and product response times. Performing a performance test ensures the product meets the required speed requirements under stress, as well as a soak test, which observes ‘whether intense and sustained activity over time shows a potential drop in performance levels, making excessive demands on the resources of the system’ (narwhal, 2018).

Functionality Test:

A functionality test ‘ensures that the software performs its intended functions correctly and meets the needs of the user’ (Bose, 2021). For this test, a product run will be conducted and filmed to ensure that the product and system work as intended. This test will use all the components within the product, for instance, the gripper, the AL5D Robotic Arm, and the shelving which was created for the balls to sit on. If the product detects there is no ball, picks up the golf ball from the shelving, and places the ball on a specified spot this will be considered as a passed functionality test as this is the intended function of the product.

Gripper Design

Design 1

Low-Level Design:

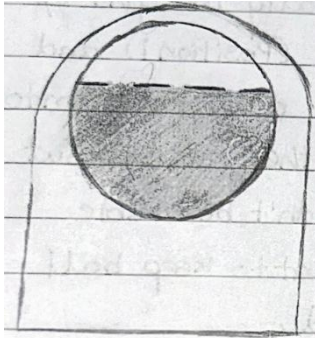


Figure 1. Low-level birds-eye view.

- Retractable base.
- Radius about 5mm larger than the dimensions of a golf ball.
- Ball enters from above (with a possible release mechanism built into the shelving, which could be a pressure point or magnet).
- Could have a possibility of around 3 different bases to release the ball. These are outlined below.

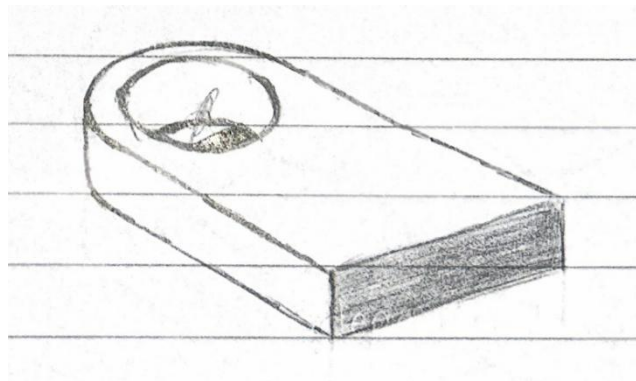


Figure 2. Low-level trimetric view

Retractable base designs:

4-Point:



- Can sync coordinates of the tee to centre point of the box for more accurate placement.
- Possibly won't hold the weight of the ball very well.

Trap Door:



- Less accurate but possibly easy to pull off, more likely to hold weight of ball.

Retractable:



- Retracting base would hold ball, although less accurate placement but if ball control and co-ordinates correct could work well (simple but effective).

High-Level Design:

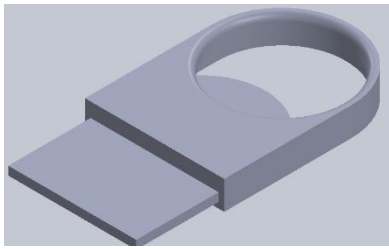


Figure 4. Trimetric view.

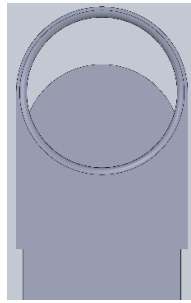


Figure 5. Birds-eye view.

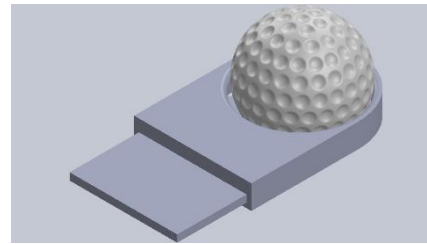


Figure 3. Trimetric view with golf ball.

Design 2

Low-Level Design:

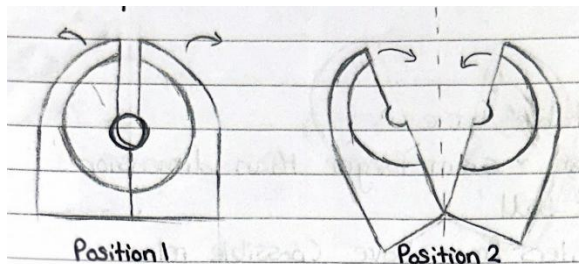


Figure 6. Low-level birds-eye view of positions 1 and 2.

- Initial idea was rigid and stuck in Position 1, to have the robot lower onto tee then reverse, but that wouldn't be stable.
- Indent used to keep the ball steady.

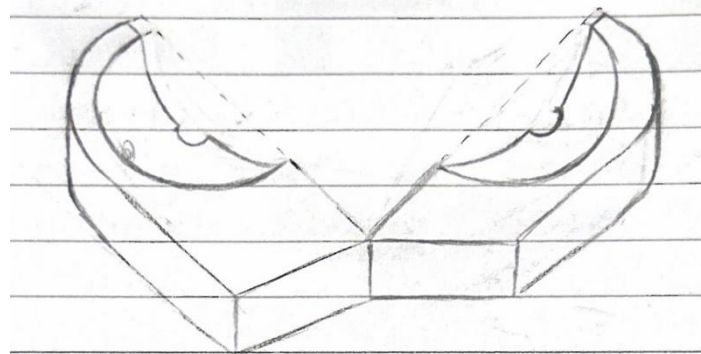


Figure 7. Low-level trimetric view of position 2.

- Can be used for x-axis motion, but not y-axis.
- Would be suitable if team agrees to move in the x-axis.

For y-axis motion:

- Could be a box, with base having indent.
- Have to open the gripper double the diameter of golf ball to get golf ball in.
 - o Could be quite cumbersome as a result.

High-Level Design:

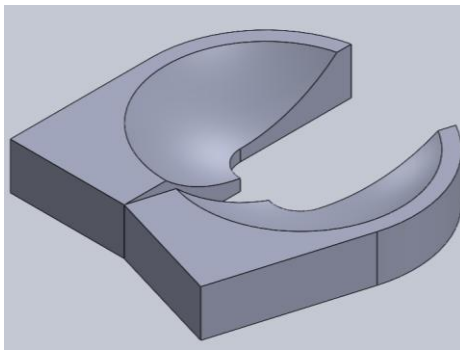


Figure 9. Trimetric view of position 2.

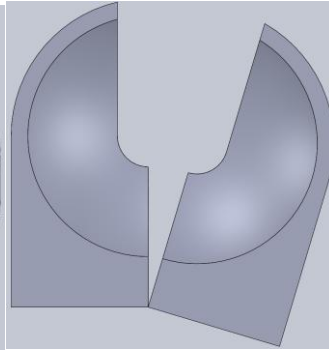


Figure 10. Birds-eye view of position 2.

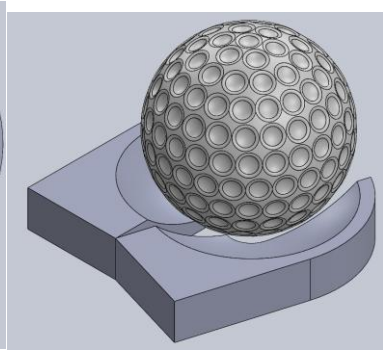
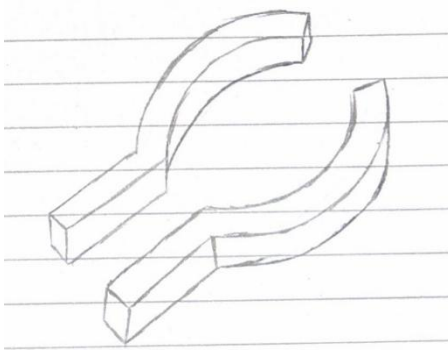


Figure 8. Trimetric view of position 2, with golf ball.

Design 3

Low-Level Design:



- Simple design with 2 curved arms, which can be moved in both the x and y axis.
- Curve radius would be the same as the radius of the golf ball.
- The design may need a non-slip material to allow for a better grip for the ball.
- The ball could be easily knocked out of the arms, and if redone, would need a third and possibly fourth arm to ensure the ball stays secure.

Figure 11. Low-level trimetric view.

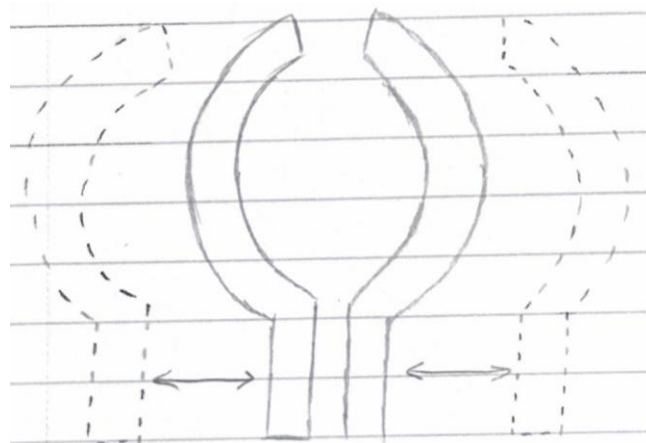


Figure 12. Low-level birds-eye view, showing motion.

High-Level Design:

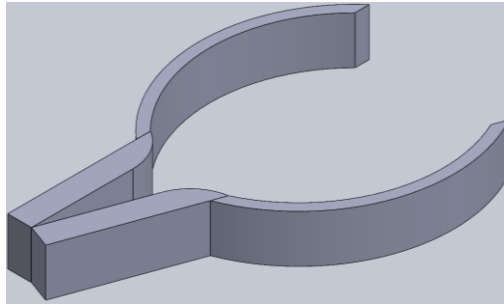


Figure 13. Trimetric view.

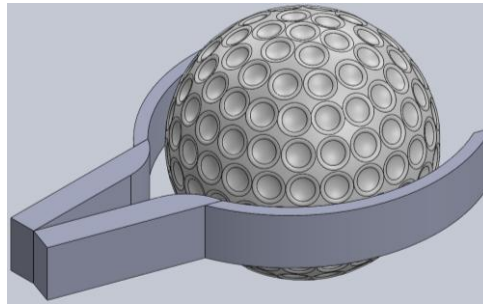


Figure 14. Trimetric view with golf ball.

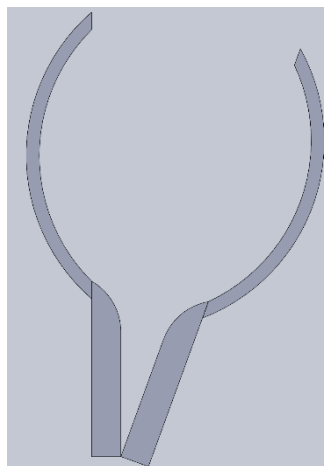


Figure 15. Birds-eye view of open position.

Design 4

Low-Level Design:

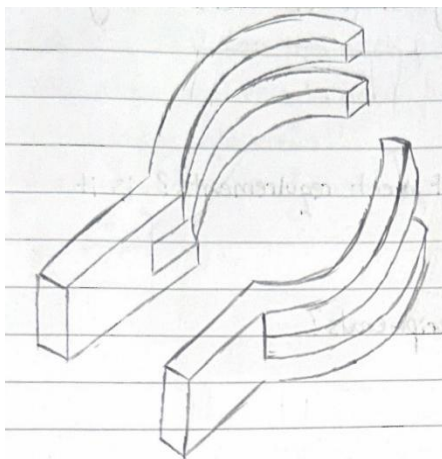


Figure 16. Low-level trimetric view.

- Two-pronged, curved, two-arm design which allows for better grip on the golf ball.
- The curve of the arm is of the same radius as the golf ball.
- This design can be used in both the x and y axis, which makes it easy for coding.
- The design may need a non-slip material to help with the grip of the ball.
- The two-prong design allows for more surface area of the ball to be covered when being picked up. Due to the spacing between the prongs, there is a less likelihood of the ball falling out of grip.

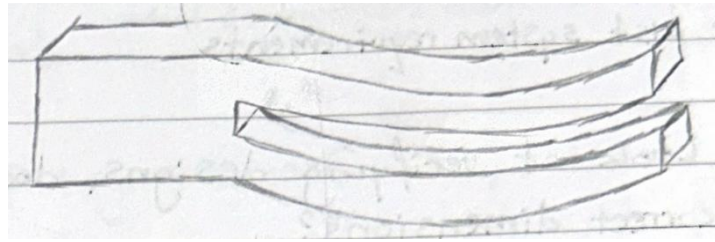


Figure 17. Side view of the gripper.

High-Level Design:

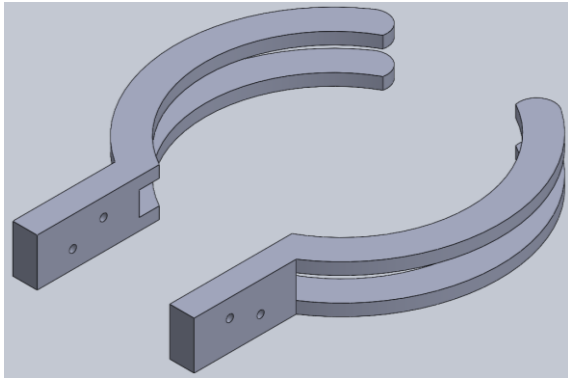


Figure 19. Trimetric view with golf ball.

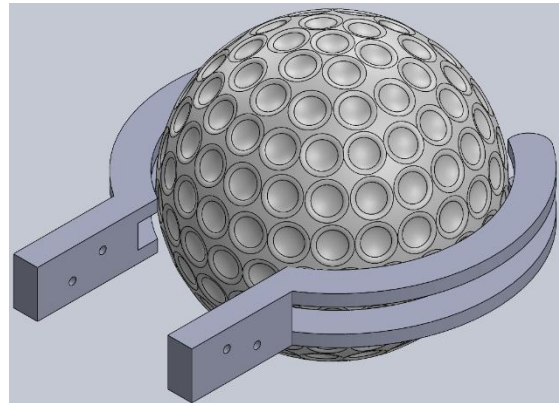


Figure 18. Trimetric view of open

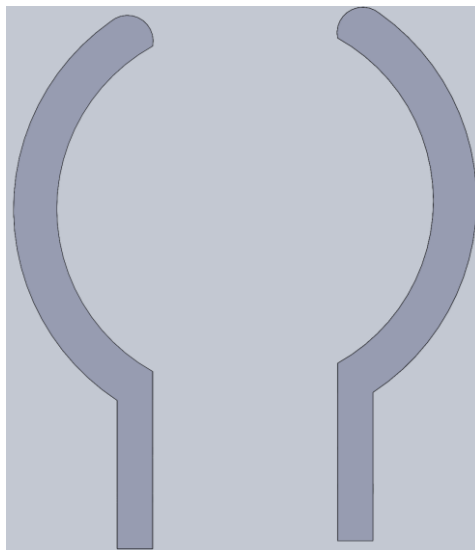
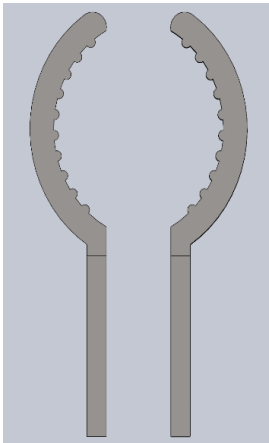


Figure 20. Birds-eye view.

Design 4 – Redesigned

High-Level Design:



- This design is a continuation of design 4 (detailed on the previous page), with some modifications to further enhance the design.
- For this design, a serrated edge was added, to improve grip of the ball.
- The two prong spacing was also increased to improve grip and control.
- The spacing was increased to ensure that the gripper was compatible with moving in both the X and Y axis.
- The length of the base of the arms was also increased to ensure that the reach met 50cm as required in the brief.

Figure 21. Birds-eye view.

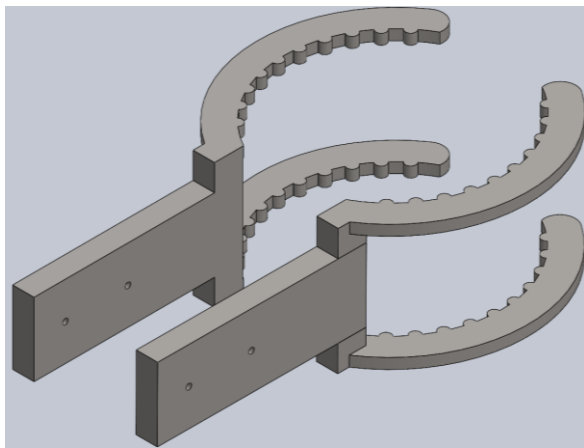


Figure 23. Trimetric view of open position.

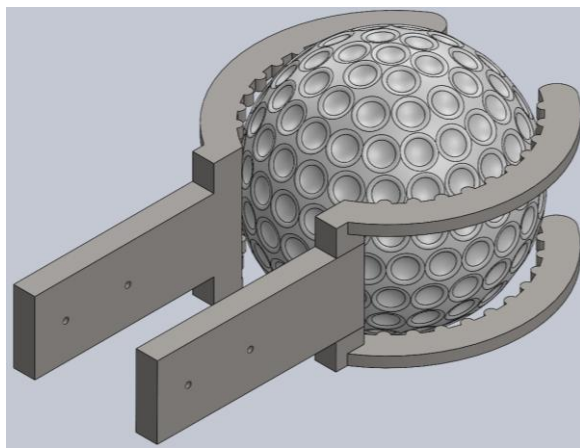


Figure 22. Trimetric view with golf ball.

Design 5

Low-Level Design:

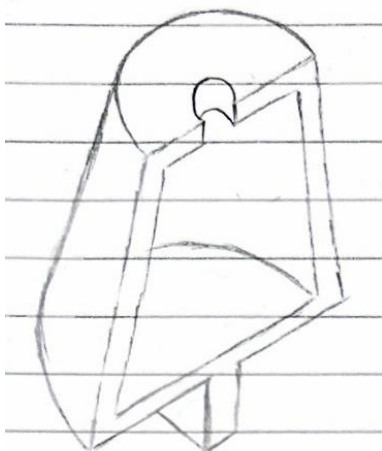


Figure 24. Low-level view of one arm.

- This design is a simple cup design, which has two arms.
- Within this design the ball is secure during all types of motion and therefore can be used in both x and y axis.
- The ball is entirely enclosed, however this may cause issues to seeing it there is a ball being carried by the robot, if a failure were to happen.
- The arms would need to open to twice the diameter of the ball and therefore would be quite cumbersome.
- This design, although very secure is not aesthetically pleasing and would not be suitable for our customers.

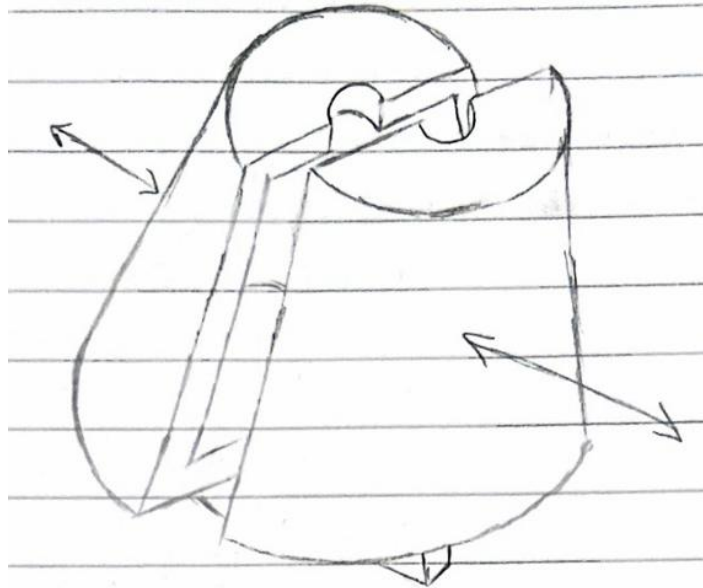


Figure 25. Low-level trimetric view of both arms, with motion indicated.

High-Level Design:

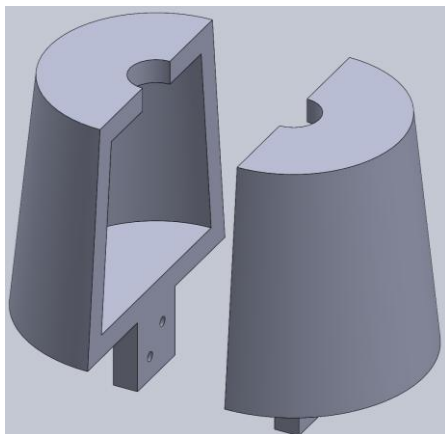


Figure 26. Trimetric view of both arms.

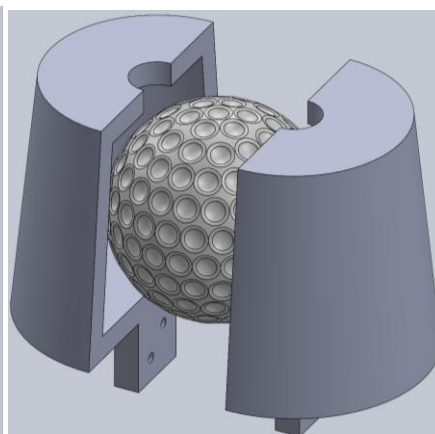


Figure 27. Trimetric view with golf ball.

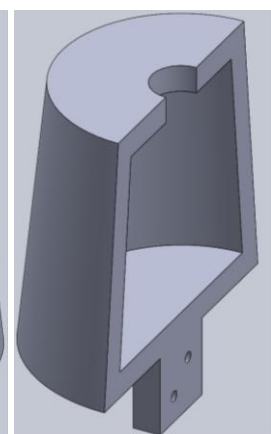


Figure 28. Single arm.

High-Level Design Discussion:

All high-level designs shown above were created in SolidWorks to scale. The dimensions which were used for the CAD models were considered when knowing the golf ball had a diameter of 43mm. This then influenced all the dimensions of the gripper designs. Another parameter which was considered when modelling the initial designs was that the robot requires an overall reach of 50cm. For this, the team decided on a simple method, of adding the newly designed gripper to the already supplied gripper on the robot. This added length to the reach, and the dimensions of the gripper were then modified to ensure the reach of the AL5D arm is 50cm. This is visible below in Figure... when the robot CAD model is discussed. As the design was decided to be added to the initial design, it was then considered on how to attach the

new design to the current attachment. When investigating the AL5D gripper further, it was found there were 2 holes present in each side, which could be used to place screws, or bolts (due to its unthreaded nature). This was then a new parameter which required holes to be created in the designs. These holes are visible in Designs 4, 4 redesigned, and 5.

An analysis was conducted for all designs shown above to ensure the correct design was selected for the task, this is visible in Table 1. The initial analysis was ranked as a result of the impact the negatives of the design would have on the project. From the rankings, it was decided that Design 4's redesign was the best for the job. A verification was then conducted to ensure that this design was also able to meet the most requirements and therefore confirm the use of the design. The gripper verification table is shown below in Table 2. The verification table shows that Design 4's redesign was the correct choice as it met more requirements and stakeholders' needs than any other design. For this project however, Design 3, and the original Design 4 came a close second as a result of only not meeting one of the requirements, which was the reliability stakeholder requirement. These two designs did not meet these criteria due to the potential for the ball to fall out of grip. Design 4 redesigned therefore had an increased likelihood of the ball not falling out of grip due to the enhanced grip methods, for example, the 2 arms spaced so the ball is supported from top and bottom, as well as the serrated edge.

High-Level Gripper Design Analysis

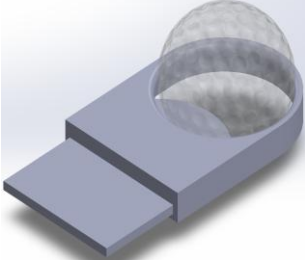
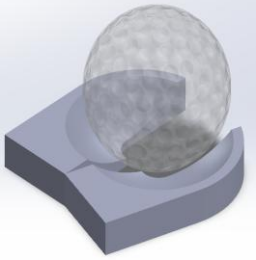
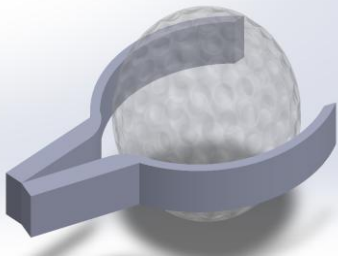
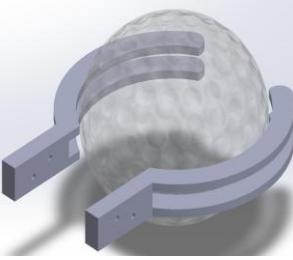
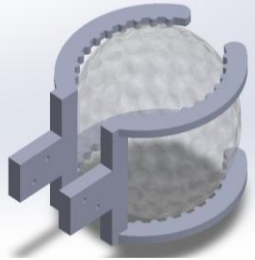
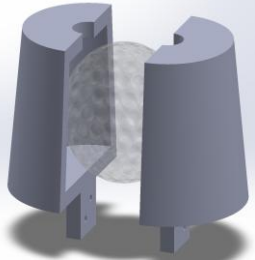
Design	1	2	3	4	4 Redesigned	5
CAD Model						
Description	Solid rectangular piece of material curved at the front, with a golf ball diameter hole. The golf ball can be positioned using a flat, retractable piece, which can be moved positively and negatively in the z-axis.	Two solid pieces of material, curved at the front, with a curved inprint to fit a golf ball. The golf ball can be positioned by the two pieces separating in the x direction. There is a channel in which the tee will slot in for accurate positioning.	Two thin arms of material curved in the x direction, connecting to straight arms. The x-axis curves allow the golf ball to be easily placed. The arms will move in the x direction to pick and place the ball.	Four metal arms, curved to the radius of the golf ball. Two arms on the side will be connected, with 2 holes in each to screw onto the originally designed gripper. The arms will move in the x direction.	A four-prong design curved to the radius of the golf ball with serrated edges for better grip. The two arms are separated at a distance of 20mm to allow the ball to sit mostly contained in the prongs. There are 2 holes to screw onto original gripper.	Two-section fully contained cup design, curved to the dimensions of the golf ball. The two sides close together in the x direction, for lifting in the y direction. There is a specific hole for tee placement.
Positives	- Easily manufactured. - Low material cost. - Simple design.	- Easily manufactured. - Low material cost. - Accurate ball placement. - Easy to code.	- Easily manufactured. - Low cost. - Simple design. - Easy to code.	- Easily manufactured. - Recyclable. - Ball is held better. - More accurate ball positioning. - More durable.	- Good grip of ball. - Recyclable. - Durable.	- Accurate placement. - No risk of ball dropping. - Durable. - Recyclable. - Sustainable.
Negatives	- Golf ball isn't as secure as desired. - Difficult to code the robot to push and pull the retractable piece. - Inaccurate placement of the golf ball. - Only works in 1 direction.	- Golf ball isn't secure, could easily fall off. - Not aesthetically pleasing for a customer. - Only works in 1 direction.	- Golf ball isn't secure, can easily slip out. - Not aesthetically pleasing. - No extra grip properties. - Ball placement accurate, but not guaranteed.	- Golf ball could slip out of the grasp of the gripper. - Not aesthetically pleasing.		- No visibility for occupancy. - Has to expand double diameter of ball. - Not aesthetically pleasing. - Only works in 1 direction.
Score	5/10	7/10	6/10	8/10	9/10	8/10
Improvements	- Make the main piece deeper to ensure the safe transport of the ball. - Re-evaluate the retractable piece, and possibly make a more accurate piece for placement. - Improve aesthetics of design. - Improve sustainability, possibly reduce the amount of material used.	- Make inprint deeper, or create side walls to ensure safe transport of the golf ball. - Make more aesthetically pleasing. - Improve sustainability, reduce amount of material used. - Make recyclable.	- Add a grip material or edge to enhance grip on the ball. - Make it more aesthetically pleasing. - Ensure recyclable material is used. - Make arms wider, to increase surface area for maximum grip. - Curve in the Y direction to enhance grip.	- Add a grip material or a serrated edge to provide better hold. - Ensure the position of the screw holes and size of screw holes are correct. - Ensure no sharp edges are present, as made of metal and could cut consumers.	- Metal could be polished, to make design more aesthetically pleasing.	- Use transparent material for visibility of the ball. - Use a gear system for wider opening, but less space used. - Improve aesthetics.

Table 1. Gripper Design Analysis

Gripper Design Verification

ID	Stakeholder Requirement	Design 1	Design 2	Design 3	Design 4	Design 4.2	Design 5		Key:
ST1	Safe to operate	M	M	M	M	M	M	M	Meets
ST2	Place golf balls on the tee within 5 seconds	M	M	M	M	M	M	DM	Doesn't meet
ST3	Have a 'teach-mode'	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Not Applicable
ST4	Have a reach of 50cm	N/A	N/A	N/A	N/A	N/A	N/A		
ST5	Low cost for system setup and maintenance	M	M	M	M	M	M		
ST6	Easy to replace the components	M	M	M	M	M	M		
ST7	Cheap and easy to purchase spare parts	DM	DM	M	M	M	DM		
ST9	Profitable	DM	DM	M	M	M	DM		
ST10	Easy to use	DM	DM	M	M	M	DM		
ST11	Reliable	DM	DM	DM	DM	M	DM		
ST12	Optional outdoor use	M	M	M	M	M	M		
ST13	Should not cause any harm to user	M	M	M	M	M	M		
ST14	Peripherals and gripper must be sustainable	M	M	M	M	M	M		
ST15	Must include emergency stop and failsafe mechanism	N/A	N/A	N/A	N/A	N/A	N/A		
ST16	Shelf must contain 50 balls	N/A	N/A	N/A	N/A	N/A	N/A		
ID	System Requirement	Design 1	Design 2	Design 3	Design 4	Design 4.2	Design 5		
SR1	Must be able to pick up a golf ball of radius 43mm	M	M	M	M	M	M		
SR2	Must be able to place a golf ball on specific point, of radius 11mm	DM	M	M	M	M	M		
SR3	Gripper must remain level during operation	M	M	M	M	M	M		
SR5	Inexpensive to manufacture	DM	M	M	M	M	DM		
SR6	Safe to manufacture	M	M	M	M	M	M		
SR7	Must be able to operate from safe distance	M	M	M	M	M	M		
SR8	Must be user-friendly	DM	DM	M	M	M	DM		
SR9	Corrosion resistance	M	M	M	M	M	M		
SR10	Must maintain functionality in temperatures ranging from -5°C to 40°C	M	M	M	M	M	M		
SR11	Must maintain functionality in light rain conditions (IPX4 water resistance)	M	M	M	M	M	M		
SR12	Must have teach-mode function	N/A	N/A	N/A	N/A	N/A	N/A		
SR13	The robot must move faster than placing by hand	N/A	N/A	N/A	N/A	N/A	N/A		
SR14	Must have the option for tee or no tee	M	DM	M	M	M	DM		
SR15	Must contain camera	N/A	N/A	N/A	N/A	N/A	N/A		
SR16	Robot operation must stop if obstacle is detected	N/A	N/A	N/A	N/A	N/A	N/A		
ID	Technical Requirement	Design 1	Design 2	Design 3	Design 4	Design 4.2	Design 5		
TR1	Gripper must fit within the original claw	M	M	M	M	M	M		
TR2	Redesign gripper to be able to fit standard golf ball size	M	M	M	M	M	M		
TR3	Must have a delay of 3 seconds	N/A	N/A	N/A	N/A	N/A	N/A		
TR4	Must be able to release one ball at a time on command	M	M	M	M	M	M		
TR5	Must have a webcam	N/A	N/A	N/A	N/A	N/A	N/A		
TR6	Must have sensors	N/A	N/A	N/A	N/A	N/A	N/A		
TR7	Increase the robot arm length to reach the desired point	N/A	N/A	N/A	N/A	N/A	N/A		
Amount of Requirements Met:		18	19	24	24	25	18		

Table 2. Gripper verification using requirements.

Gripper Manufacturing

Gripper Manufacturing Considerations

Manufacturing Process	3D Printing	CNC Machining and Wire Bending	Injection Moulding	Thermoforming
Description	3D printing is an additive manufacturing process that creates an object from a CAD model by laying down thin layers of a material and then fusing the layers together.	CNC cutting involves the use of computer-controlled cutting machine, to accurately cut materials with high precision and intricate detailing. Wire bending is the process used to bend a plastics, by using a heated wire.	Injection moulding is a manufacturing process that involves injecting molten plastic into a mould and then cooling and solidifying it. The molten plastic is injected into the mould at high pressure, before cooling and taking on the desired shape.	Thermoforming is a manufacturing process that involves heating a plastic sheet until it's pliable, then forming it into a shape using a mold, and colling it to retain the shape.
Average Cost/hour	£15	£60 (when being made at supplier)	£100 (when being made at supplier)	£500 (being made at supplier and considering custom mold cost)
Average Time	1 hour (could be more depending on size and complexity of the design)	1 hour (could be less depending on the rpm of the machine)	2 minutes (could be more depending on the cooling time for the material)	15-20 hours (depending on design complexity)
Advantages	- Flexibility for design. - Rapid Prototyping. - Strong and Lightweight parts. - Minimal waste. - Environmentally friendly. - Cost effective.	- Unparalleled precision. - Consistency. - Increased efficiency. - Complex design capabilities. - Enhanced safety.	- Allows complex geometries with tight tolerances. - Efficient. - High repeatability and reliability. - Can reuse material.	- Can be used for both large and small items. - Suitable for most types of plastic. - Simple piece of manufacturing to learn. - High quality products produced in short period.
Disadvantages	- Limited material availability. - Parts require clean up post processing. - Layers can delaminate under strain. - Restricted build size.	- High initial cost. - Requires skilled labour. - Material limitations. - Complex maintenance requirements. - Inefficient for small production runs.	- High start-up costs. - Long initial lead times. - Design changes are expensive.	- Restricted to thin-walled designs. - More costly. - Restricted to plastic sheets. - Thickness of part can vary.
Score Based Off Requirements	77.78%	77.78	77.78%	55.56%

Table 3. Manufacturing considerations.

Manufacturing Technique Verification

ID	Stakeholder Requirement	3D Printing	CNC Machining	Injection Moulding	Thermoforming
ST1	Safe to operate	N/A	N/A	N/A	N/A
ST2	Place golf balls on the tee within 5 seconds	N/A	N/A	N/A	N/A
ST3	Have a 'teach-mode'	N/A	N/A	N/A	N/A
ST4	Have a reach of 50cm	N/A	N/A	N/A	N/A
ST5	Low cost for system setup and maintenance	M	DM	DM	DM
ST6	Easy to replace the components	M	M	M	M
ST7	Cheap and easy to purchase spare parts	M	M	M	M
ST9	Profitable	M	M	M	DM
ST10	Easy to use	N/A	N/A	N/A	N/A
ST11	Reliable	DM	M	M	M
ST12	Optional outdoor use	N/A	N/A	N/A	N/A
ST13	Should not cause any harm to user	N/A	N/A	N/A	N/A
ST14	Peripherals and gripper must be sustainable	M	M	M	DM
ST15	Must include emergency stop and failsafe mechanism	N/A	N/A	N/A	N/A
ST16	Shelf must contain 50 balls	N/A	N/A	N/A	N/A

ID	System Requirement	3D Printing	CNC Machining	Injection Moulding	Thermoforming
SR1	Must be able to pick up a golf ball of radius 43mm	N/A	N/A	N/A	N/A
SR2	Must be able to place a golf ball on specific point, of radius 11mm	N/A	N/A	N/A	N/A
SR3	Gripper must remain level during operation	N/A	N/A	N/A	N/A
SR5	Inexpensive to manufacture	M	M	DM	DM
SR6	Safe to manufacture	M	M	M	M
SR7	Must be able to operate from safe distance	N/A	N/A	N/A	N/A
SR8	Must be user-friendly	N/A	N/A	N/A	N/A
SR9	Corrosion resistance	DM	DM	M	M
SR10	Must maintain functionality in temperatures ranging from -5°C to 40°C	N/A	N/A	N/A	N/A
SR11	Must maintain functionality in light rain conditions (IPX4 water resistance)	N/A	N/A	N/A	N/A
SR12	Must have teach-mode function	N/A	N/A	N/A	N/A
SR13	The robot must move faster than placing by hand	N/A	N/A	N/A	N/A
SR14	Must have the option for tee or no tee	N/A	N/A	N/A	N/A
SR15	Must contain camera	N/A	N/A	N/A	N/A
SR16	Robot operation must stop if obstacle is detected	N/A	N/A	N/A	N/A

ID	Technical Requirement	3D Printing	CNC Machining	Injection Moulding	Thermoforming
TR1	Gripper must fit within the original claw	N/A	N/A	N/A	N/A
TR2	Redesign gripper to be able to fit standard golf ball size	N/A	N/A	N/A	N/A
TR3	Must have a delay of 3 seconds	N/A	N/A	N/A	N/A
TR4	Must be able to release one ball at a time on command	N/A	N/A	N/A	N/A
TR5	Must have a webcam	N/A	N/A	N/A	N/A
TR6	Must have sensors	N/A	N/A	N/A	N/A
TR7	Increase the robot arm length to reach the desired point	N/A	N/A	N/A	N/A
Amount of Requirements Met:		7	7	7	5

Table 4. Manufacturing technique verification.

When considering the gripper design, it was also important to consider the ideal manufacturing technique which would be used for production. This method must be compatible with the stakeholder requirements. This can be seen in Table 4 below, which uses the same key as in Table 2. When considering manufacturing techniques, it must be noted the 'best' technique is not always feasible for a start-up or a prototype creation, nor is it always feasible for large batch productions, when a product is in high demand. Table 3 shows that 3D printing was the best manufacturing procedure for the gripper, however, it is also important to remember

the disadvantages listed would not be feasible in large batch production. Having parts requiring post-processing clean-up will mean more time is needed to roll out the product, as well as more employees required to complete this job, and for quality assurance. Although this is a minor issue and provides employment opportunities for local communities, this is a costly requirement for a start-up business and may not be able until a budget is available for hiring within these positions. In the future, it would be ideal to use injection moulding as the manufacturing technique as it takes the least time, and fewer people are required for the manufacturing to run smoothly. This method is costly for start-up, however, once the initial set-up costs are covered, it is relatively low cost to run. For the prototype, 3D printing was chosen as the favoured technique. This meant a single run could be completed, for a relatively low cost, as well as in a fast manner, to allow other verification and validation techniques to be completed for the deadline. There was a restriction on what material could be used, due to the supplier however the topic of material is discussed further on in the document.

After considering the manufacturing process, it is important to consider the materials that could be used when 3d printing the gripper prototype. There were 4 different materials considered, these are PLA, stainless steel, ABS, and polypropylene. When researching these materials there was a specific focus on which materials were the most sustainable, and environmentally friendly. It is important to consider the environmental impact the product would have before manufacturing, to try and reduce the harm caused from the beginning. It was found that PLA and stainless steel had the least environmental impact, due to ABS and polypropylene both being created by fossil fuels and therefore not having renewable sources. Although PLA is more renewable than ABS and polypropylene, it is not the best material due to being the recyclability of the material is also an important fact to consider, stainless steel has the best properties for recycling as the overall properties of the material are not hindered by the recycling process. This allows the product to be more environmentally friendly as there would be less landfill waste caused by this product.

When verifying the materials against the requirements for the product, it was found that the PLA and ABS met the requirements the most. ABS however did not meet the sustainability requirement, which is arguably the most important requirement. This means that the PLA is the best material for the prototype. However, PLA is also flawed as it does not meet the optional outdoor use requirement, as well as corrosion resistance. PLA does not meet on the outdoor use requirement as it doesn't withstand sunlight well, which can cause the plastic filament to be brittle and break.

3D Printing Material Considerations

Materials	PLA	Stainless Steel	ABS	Polypropylene
Description	Polylactic Acid (PLA) is one of the most popular materials used in desktop 3D printing. It is the default filament of choice for most extrusion-based 3D printers as it can be printed at a low temperature and does not require a heated bed.	Steels are 3D printed using the DMLS (Direct Metal Laser Sintering) or SLM process. A very fine metal powder is melted with a laser to produce your design layer by layer. Once the design is complete, support structures are removed.	ABS (Acrylonitrile Butadiene Styrene) is a thermoplastic which becomes pliable when heated and returns to solid when cooled. ABS is 3D printed onto a heated bed with a temperature of around 100 degrees to prevent warping and improve adhesion to the platform.	Polypropylene is a semi-crystalline material which requires a precise temperature through the printing process. It is used mostly in FDM and SLS applications. These applications take advantage of the material's lightweight nature and fatigue resistance.
Advantages	- Low cost. - Stiff and good strength. - Good dimensional accuracy. - Good shelf life.	- Rapid prototyping. - Cost-effective customisation. - Excellent corrosion resistance. - Fatigue resistance for cyclical loading.	- Strong and durable. - Chemically resistant. - Dimensional stability. - Medium heat tolerance.	- Chemical resistance. - Flexible and tough. - Lightweight. - Smooth surface finish.
Disadvantages	- Low heat resistance. - Can ooze and may need cooling fans. - Filament can get brittle and break. - Not suitable for outdoors.	- Prone to warping. - High-quality metal powders are expensive. - Support structure removal can leave marks.	- Causes strong odours when printing. - Prone to warping during printing. - Difficult to post process (requires acetone for a smooth finish).	- High printing temperature. - Expensive. - Sensitive to UV light and oxidations, making it not suitable for outdoor use. - Warping.
Environmental Impact	Derived from plant-based sources, so more environmentally friendly compared to ABS. But it depletes its natural resources faster than it can be replenished. This material is biodegradable within high heat industrial recycling centres.	Stainless steel is a recyclable and sustainable material, due to its capability to be recycled infinitely without reducing quality. During stainless steels resolidification restores metallic bonds, which restores the original properties of the metal.	Due to ABS being derived from petrochemicals, it is a non-renewable material. The production of the materials relies on fossil fuel extraction, which contributes to carbon emissions. ABS however can be recycled, but not easily.	Manufacturing polypropylene has a high impact on fossil fuel depletion and global warming. The process is a significant contributor to greenhouse gas emissions. Polypropylene also releases toxins when burnt, resulting in a severe environmental threat.
Score	84.62%	76.92%	84.62%	53.85%

Table 5. 3D printing materials.

Material Verification

ID	Stakeholder Requirement	PLA	Stainless Steel	ABS	Polypropylene
ST1	Safe to operate	N/A	N/A	N/A	N/A
ST2	Place golf balls on the tee within 5 seconds	N/A	N/A	N/A	N/A
ST3	Have a 'teach-mode'	N/A	N/A	N/A	N/A
ST4	Have a reach of 50cm	N/A	N/A	N/A	N/A
ST5	Low cost for system setup and maintenance	M	DM	M	DM
ST6	Easy to replace the components	M	M	M	M
ST7	Cheap and easy to purchase spare parts	M	DM	M	DM
ST9	Profitable	M	M	M	DM
ST10	Easy to use	N/A	N/A	N/A	N/A
ST11	Reliable	M	M	M	M
ST12	Optional outdoor use	DM	M	M	DM
ST13	Should not cause any harm to user	M	M	M	M
ST14	Peripherals and gripper must be sustainable	M	M	DM	DM
ST15	Must include emergency stop and failsafe mechanism	N/A	N/A	N/A	N/A
ST16	Shelf must contain 50 balls	N/A	N/A	N/A	N/A
ID	System Requirement	PLA	Stainless Steel	ABS	Polypropylene
SR1	Must be able to pick up a golf ball of radius 43mm	N/A	N/A	N/A	N/A
SR2	Must be able to place a golf ball on specific point, of radius 11mm	N/A	N/A	N/A	N/A
SR3	Gripper must remain level during operation	N/A	N/A	N/A	N/A
SR5	Inexpensive to manufacture	M	DM	M	DM
SR6	Safe to manufacture	M	M	DM	M
SR7	Must be able to operate from safe distance	N/A	N/A	N/A	N/A
SR8	Must be user-friendly	N/A	N/A	N/A	N/A
SR9	Corrosion resistance	DM	M	M	M
SR10	Must maintain functionality in temperatures ranging from -5°C to 40°C	M	M	M	M
SR11	Must maintain functionality in light rain conditions (IPX4 water resistance)	M	M	M	M
SR12	Must have teach-mode function	N/A	N/A	N/A	N/A
SR13	The robot must move faster than placing by hand	N/A	N/A	N/A	N/A
SR14	Must have the option for tee or no tee	N/A	N/A	N/A	N/A
SR15	Must contain camera	N/A	N/A	N/A	N/A
SR16	Robot operation must stop if obstacle is detected	N/A	N/A	N/A	N/A
ID	Technical Requirement	PLA	Stainless Steel	ABS	Polypropylene
TR1	Gripper must fit within the original claw	N/A	N/A	N/A	N/A
TR2	Redesign gripper to be able to fit standard golf ball size	N/A	N/A	N/A	N/A
TR3	Must have a delay of 3 seconds	N/A	N/A	N/A	N/A
TR4	Must be able to release one ball at a time on command	N/A	N/A	N/A	N/A
TR5	Must have a webcam	N/A	N/A	N/A	N/A
TR6	Must have sensors	N/A	N/A	N/A	N/A
TR7	Increase the robot arm length to reach the desired point	N/A	N/A	N/A	N/A
Amount of Requirements Met:		11	10	11	7

Table 6. Material verification.

Robot CAD Model

After verifying that all parts of the gripper design were considered and thought through, it was important to verify that the reach did in fact meet the 50cm as specified in the project brief. To do this, a CAD model of the robotic arm was created. This model was

created to scale, lengthwise to ensure that the reach is in fact correct and meets the specification before 3D printing the design. This CAD model was also created as a backup plan to ensure that the motion caused by the code was correct in case the robot was not able to be tested before the deadline.

The below figure shows the robotic arm CAD model, which was created in SolidWorks, using multiple different parts, and boxes to replicate the servos.

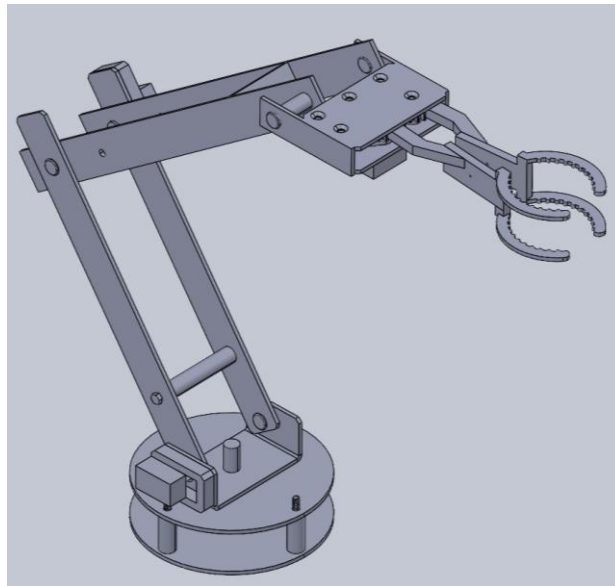


Figure 29. Simplified CAD model of the AL5D Robotic Arm.

The figure below shows that the reach does meet 50cm with the gripper design attached to the old gripper. This has been demonstrated using the measure feature in the evaluate section of the assembly. The figure below shows that the arms from the left side of the base to the tip of the gripper is 518mm which surpasses the requirement for a 50cm reach. This meets the criteria and therefore verifies the gripper with the robotic arm. In the future, if this were to be redone, I would ensure the measurement was taken from the centre of the base to the top of the gripper. This would be a more reliable measurement for the reach although it is difficult to manipulate the arm to be completely flat on the software, so this measurement would not be as accurate.

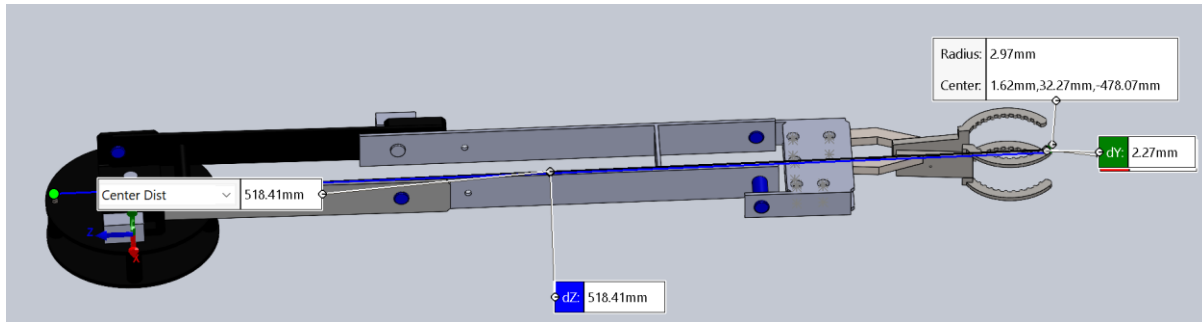
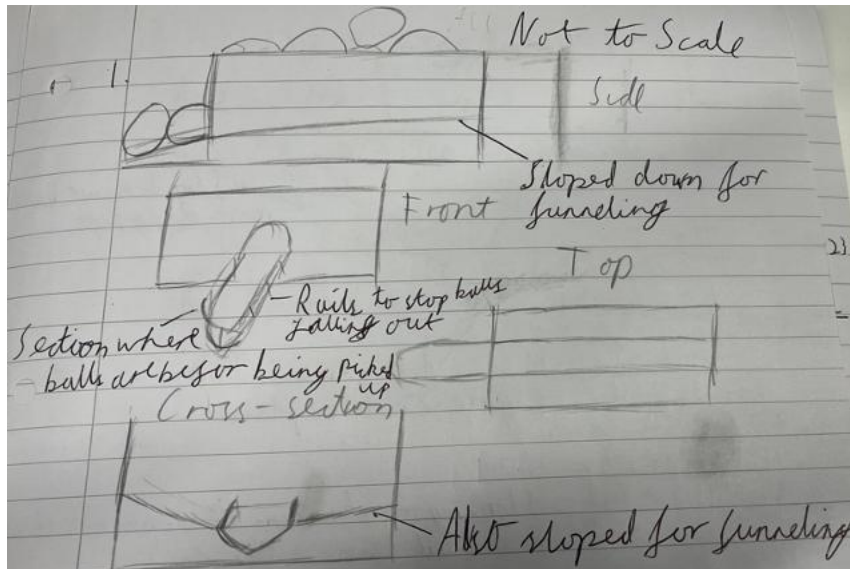


Figure 30. Robotic arm flat in the x axis, with measurements shown.

Peripheral Design

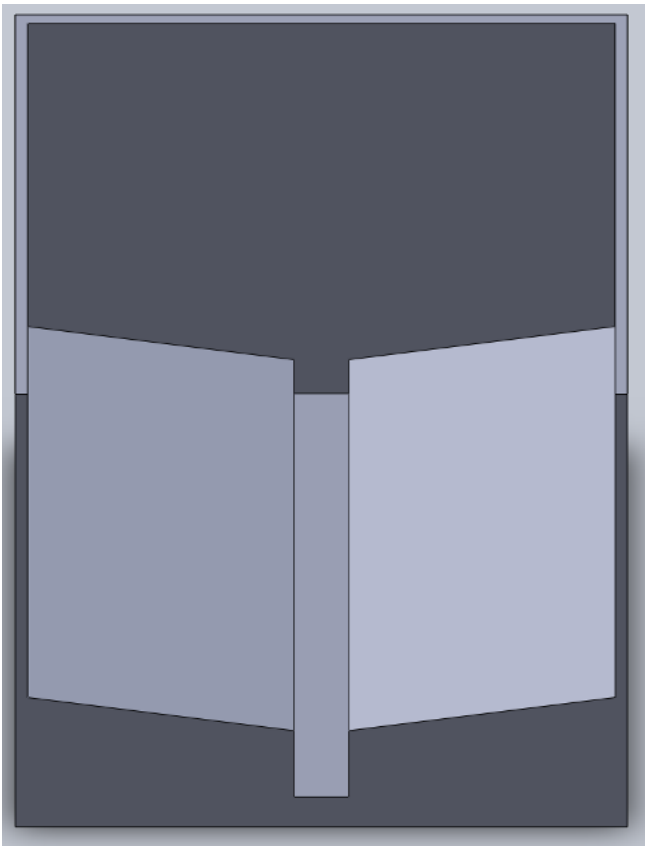
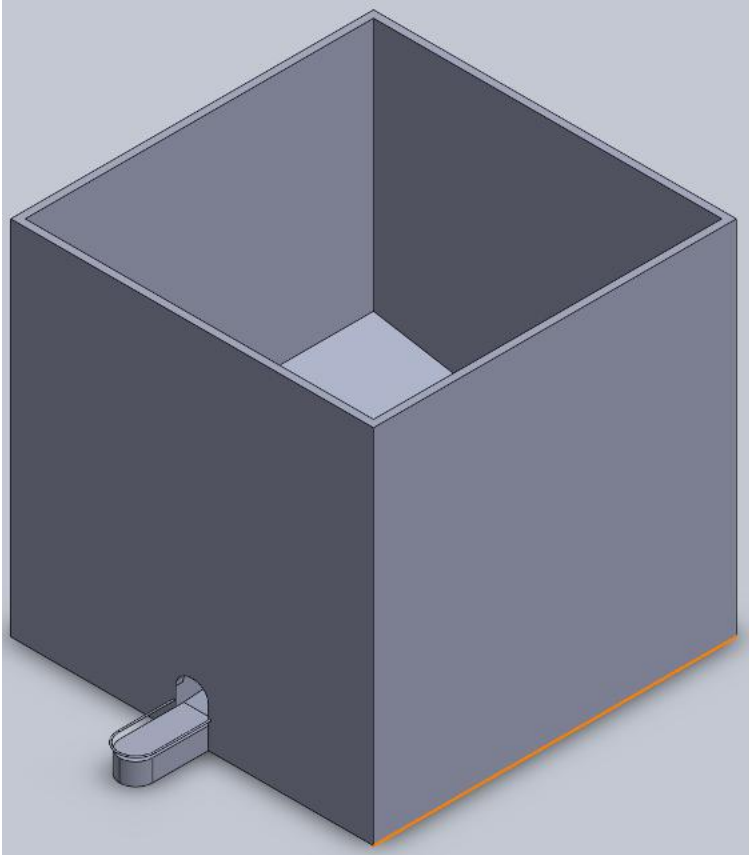
Design 1:

Low Level Design:



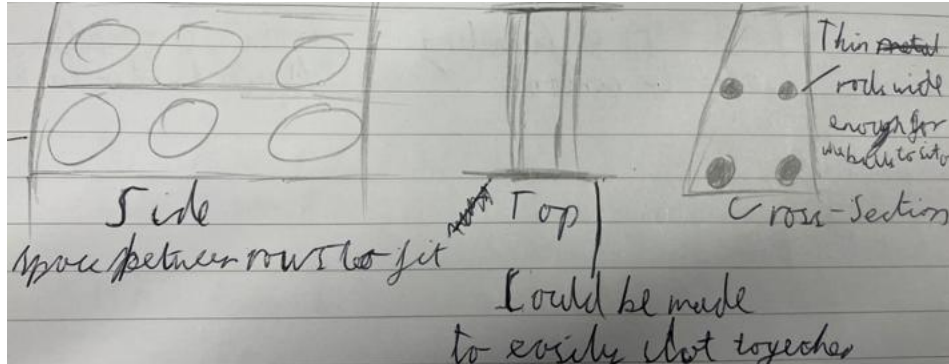
- Balls are placed in a large container with a hole going through the bottom.
- Sections would be slanted to allow balls to roll down to the hole.
- The ball would then come out of the side into a catchment area where the gripper would grab the ball.

High Level Design:



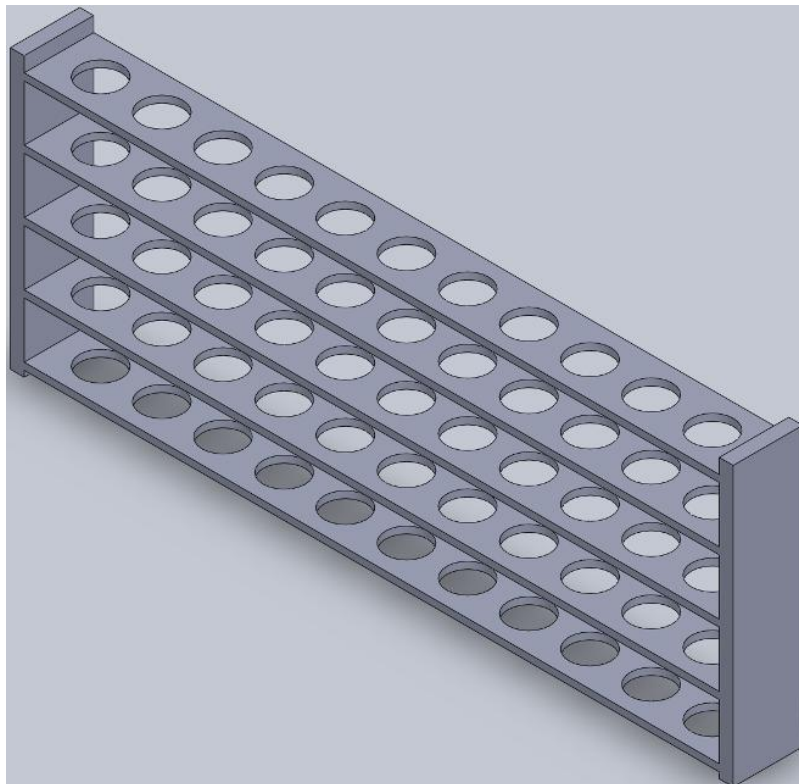
Design 2:

Low Level Design:



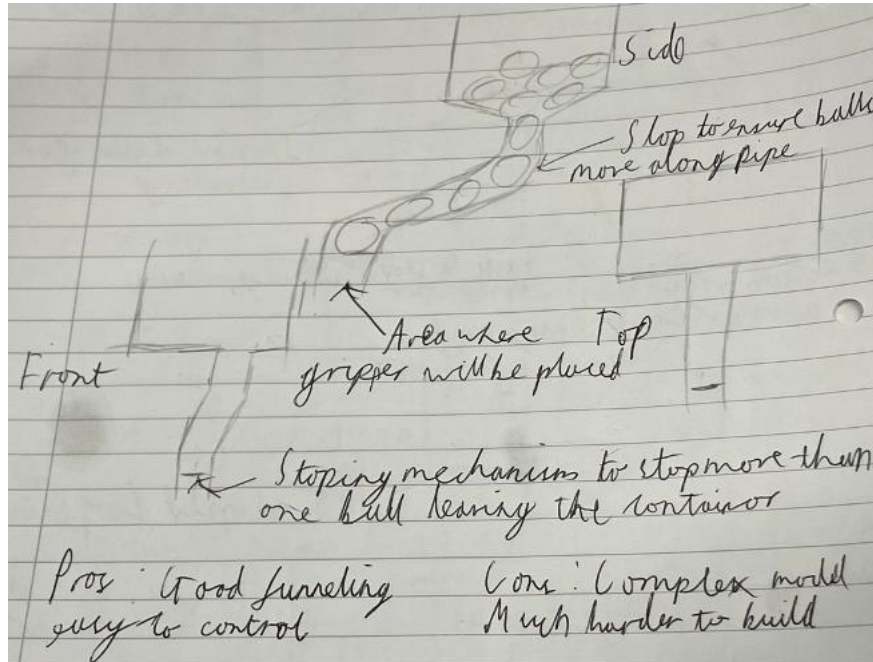
- Balls would be placed on rails so that more balls can fit.
- The grabber would then take one of these balls when it wants.
- The rack can be expanded to include multiple layers to expand storage.
- Rails can be slotted into drilled holes.

High Level Design:



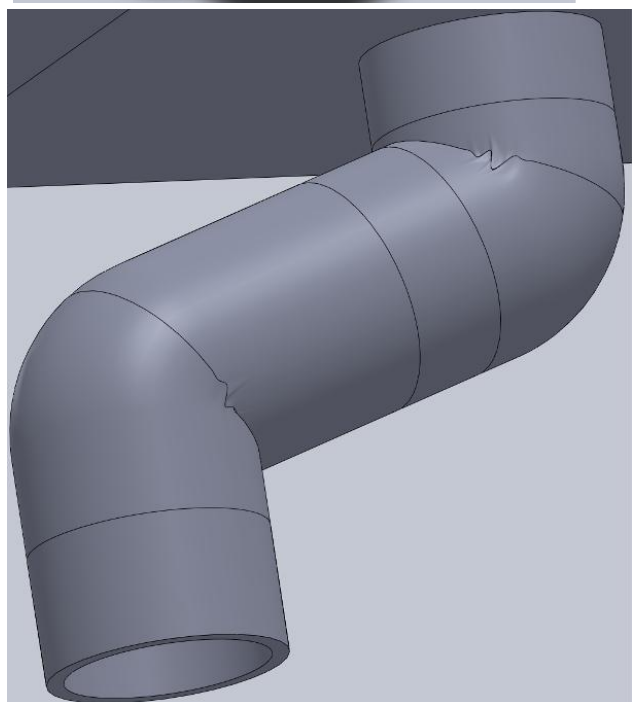
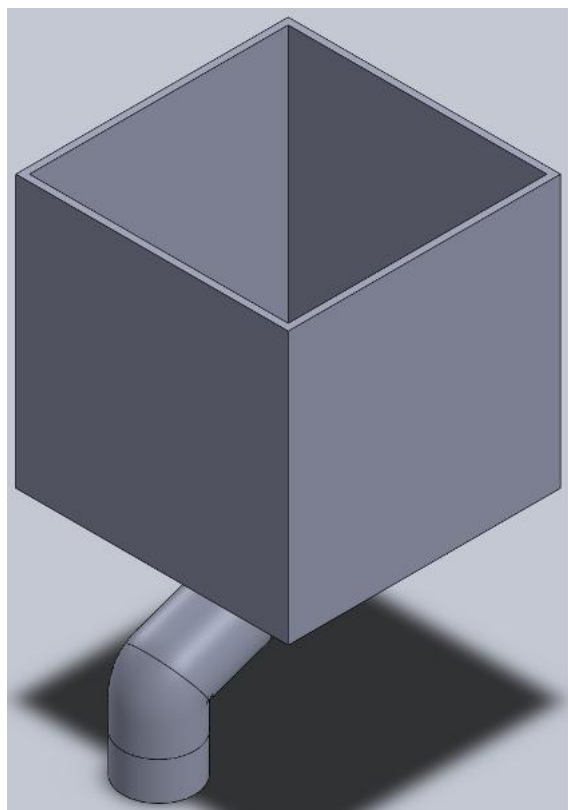
Design 3:

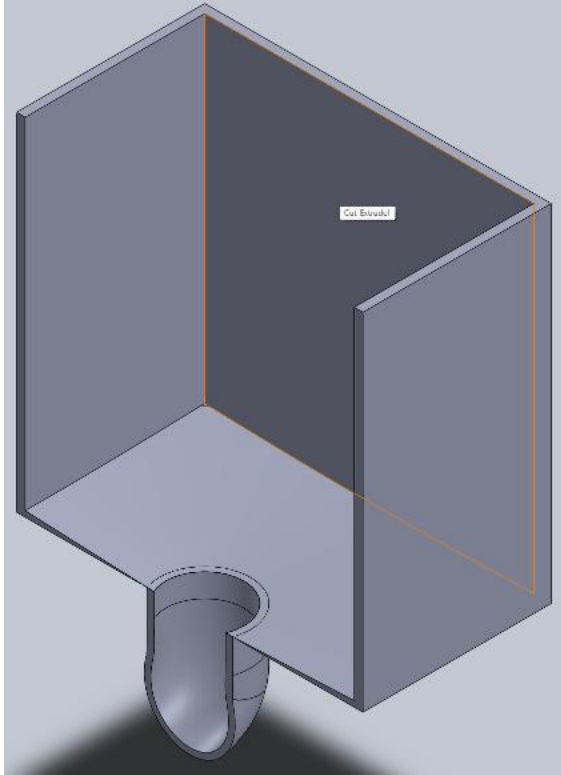
Low Level Design:



- Like Design 1 except the ball roll through a hollow tube
- A release mechanism would be in place to stop more than one ball from dropping through.
- Potential modifications can be made to stop the ball from dropping and instead roll to a specific location like in Design 1.

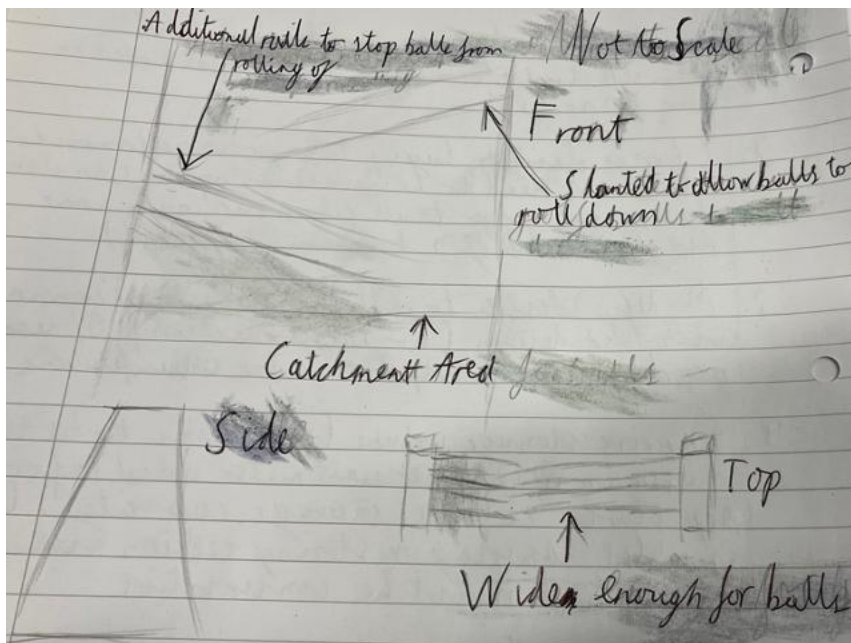
High Level Design:





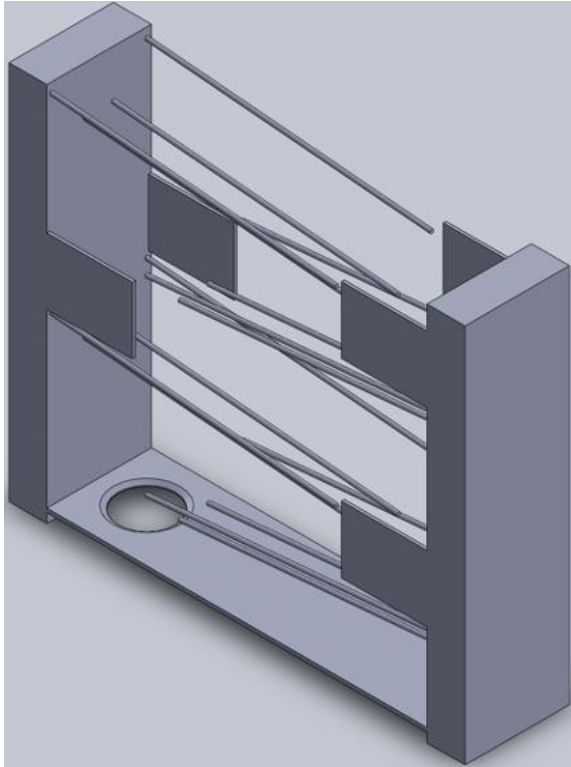
Design 4:

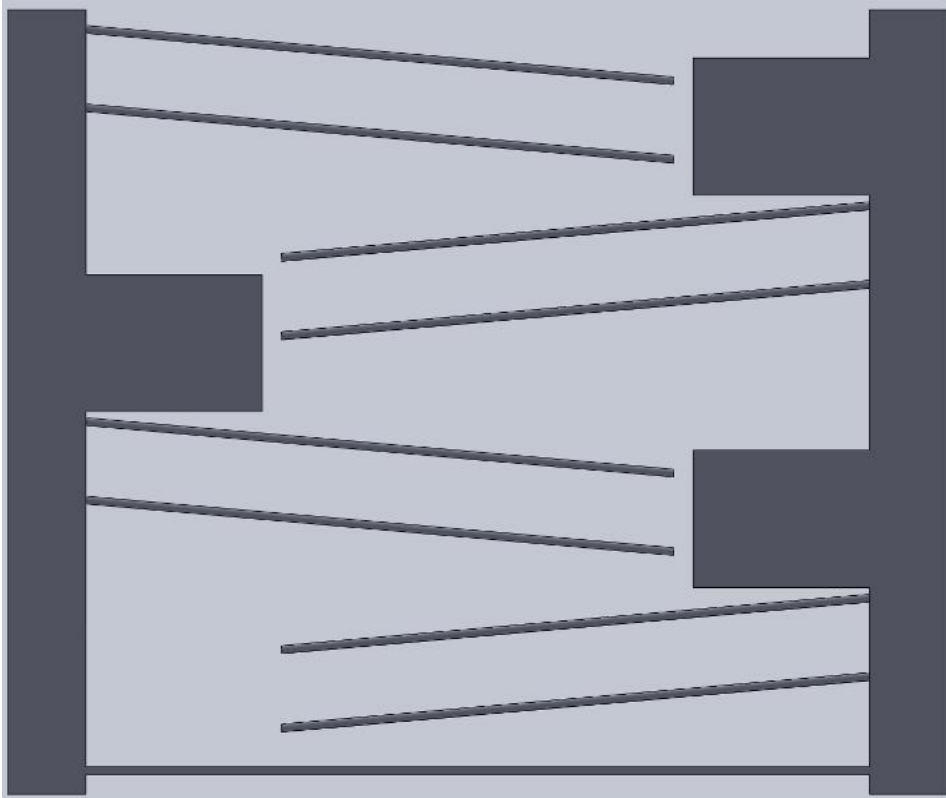
Low Level Design:



- Balls can be placed at on the top rail and roll down to the bottom.
- They then are held in a small hole until they are ready to be use.
- Rails may need to be drilled into the side of the supports much like Design 2.
- May need additional barriers where the balls drop to prevent them falling out.

High Level Design:





High-Level Design Discussion:

Much like the griper all models were design in SolidWorks to scale. All CAD models were made with a golf ball in mind (43mm) though most were built for minimum 45mm. All models were design in tandem with that of the gripers, so some designs were made with certain ones in mind. An example being design 3. Design 3 was made due to some of the earlier designs for the griper allowed the ball to be dropped onto it and carry it around before opening up and placing it. As such if it were to be used with the design, we actually went with it would require a redesign to be available from a specific point. It was also a very complex design, with the file being 4 times larger than the others. Some of the designs were biased on other storage areas in sports. Design 1 was roughly based upon a bucket while design 2 and 4 were biased on where bowling balls are normally kept, though design 2 was changed to holes to avoid highly variable ball position. In the end we ended up going for design 4 thanks to the ability so see the ball more easily and with the modification to its design, be able to store a decent number of balls within itself.

Manufacturing:

It's important to note that manufacturing a peripheral is substantially harder than the gripper. It is both more complex and much larger so it's unlikely that we would be able to use 3d printing for the whole process. Even so 3d printing can still be used for parts of the manufacturing process either for the support columns or the rails. If they aren't 3d printed then there are several alternatives that can be used, such as wood and steel. For the other panels where the ball drops down, ideally, they would be made from a transparent material to allow the user to see the balls drop down the tower.

The next major challenge would be assembling the structure. Namely the rails and the bottom platform with the hole. The most ideal solution would be to drill holes/slots into the frame and allow them to fit snugly inside. It may also be a smart idea to use an adhesive to ensure the parts stay in place, though it may also be possible to forgo any holes and simply use adhesive. The panels can either be bolted to the structure or use an adhesive to ensure that the balls are properly held since they will most likely be made from a different material.

Robotics Design

Hardware

The hardware used for this project includes a camera, an ultrasonic sensor, and the Lynxmotion AL5D robotic arm. The camera serves two purposes: identifying the pickup location of the ball and monitoring the driving zone where the ball will be dropped. This dual functionality ensures that the system can determine whether there is a ball available for pickup and verify that the player or user has played the previously dropped ball. While alternative sensors like infrared or ultrasonic sensors could also be used for detecting whether a ball is present, these sensors have specific advantages and limitations.

For detecting a ball at the pickup position, ultrasonic or infrared sensors are highly effective because the pickup position remains relatively constant. These sensors can reliably detect objects at a fixed distance without being affected by external factors, such as a person or object nearby. However, in the driving zone, where the environment is less controlled, these sensors may face challenges. For instance, an infrared sensor might detect unintended objects, leading to false positives. To address this, the system integrates an ultrasonic sensor with a camera. The ultrasonic sensor triggers the camera when it detects an object within a specific distance, allowing the camera to confirm whether the object is a golf ball. This combination ensures efficient and accurate ball detection and placement.

The hardware components mentioned above are sufficient for the current stage of prototyping. However, for future stages, a more advanced controller board will be necessary. The recommended controller for this project is an Arduino board. Arduino boards offer an abundance of input and output pins, which are essential for managing the numerous signals required for the system. These include power and control signals for the six servo motors and inputs from multiple sensors. Arduino is also cost-effective and compatible with various platforms, including Windows, macOS, and Linux (Shakirovich Ismailov, Zafar Botirovich, 2022).

In later stages, additional sensors will enhance the robot's functionality. Torque sensors can be installed on the motors to prevent damage by detecting resistance and stopping the robot if it encounters an obstacle. Proximity sensors, which are even more effective than torque sensors, can detect objects before they come into contact with the robot. This allows for improved path planning and collision avoidance. With the integration of a high-quality camera, proximity sensors, and machine learning algorithms, the robot can achieve advanced path planning. By providing the current position and the desired end position of the end effector, the system could autonomously calculate and execute an optimal path.

In summary, the hardware selection for this project balances cost, functionality, and scalability, ensuring a robust prototype while paving the way for future enhancements.

Software

MATLAB was chosen for this project because it offers a robust and versatile platform with specialized tools for robotics, image processing, machine learning, and control system design. Its **Robotics Toolbox** is particularly advantageous, providing pre-built functions for tasks like forward and inverse kinematics, trajectory generation, and control algorithm implementation. These tools streamline the development of precise and efficient robot arm operations, enabling seamless integration of motion planning and execution for the Linx Motion AL5D robot arm. The ability to quickly calculate joint angles for desired end-effector positions and generate smooth motion paths is critical for achieving accurate pick-and-place tasks. MATLAB's support for implementing advanced control strategies, such as PID controllers, further ensures the robot's precise and stable performance.

In the realm of image processing and machine learning, MATLAB stands out with its powerful capabilities for image acquisition, segmentation, feature extraction, and object recognition. These features are essential for reliably detecting golf balls in various conditions. MATLAB's machine learning tools allow for the creation of sophisticated models, such as support vector machines or neural networks, to classify images

effectively as "ball present" or "no ball." This integration of image processing and machine learning makes MATLAB an ideal choice for tasks requiring both accuracy and adaptability.

MATLAB's **Simulink environment** enhances the project by enabling simulation and prototyping. Simulink allows for the creation of virtual models of the robot system, which can be used to test and refine control algorithms in a risk-free environment. This approach reduces development time and costs by identifying potential issues before deploying the system on actual hardware. Additionally, Simulink supports rapid prototyping, allowing algorithms to transition smoothly from simulation to real-world application.

Data analysis and visualization are other areas where MATLAB excels. Its tools for data logging and visualization help monitor robot performance, analyse sensor data, and gain insights into system behaviour. This enables iterative improvements to the system's design and operation, ensuring optimal performance.

MATLAB's widespread use in academia and industry further supports its selection for this project. With extensive documentation, tutorials, and a large online community, MATLAB provides valuable resources for troubleshooting and learning. References like "Robotics, Vision & Control" by Peter Corke and "MATLAB for Robotics" by Bruno Siciliano demonstrate its application in real-world robotics research and development. By leveraging MATLAB's strengths, this project can efficiently develop, test, and deploy a reliable golf ball pick-and-place system while building industry-relevant skills.

Requirements

- User-centric design: Teach mode for customization, accommodating diverse user needs.
- Accessibility: Sound feedback for enhanced user experience and reduced eye strain.
- Safety: Emergency stop function for critical situations.
- Functionality: Meeting the 50-meter reach requirement through arm extension and a suitable gripper.

To cater to diverse user preferences and abilities, our program includes a user-friendly *teach mode*. This feature allows users to pre-program the robot's movement path, ensuring adaptability for both left-handed and right-handed users. Additionally, it supports the use of various ball dispensers, enhancing the system's versatility. By enabling users to customize the robot's operation according to their specific needs, this mode significantly improves accessibility and usability.

To further enhance the user experience and minimize visual strain, we integrated sound output as an intuitive feedback mechanism. During teach mode, auditory signals indicate robot movements and recording activities, providing users with clear and immediate feedback without relying solely on visual cues. This thoughtful design element ensures that users can interact with the system comfortably and effectively, even during extended periods of operation.

For safety, the system incorporates an emergency stop function, which can be activated by pressing the 'E' key. This feature allows users to immediately halt the robot's movements during gameplay in the event of malfunctions or unexpected issues. The emergency stop function is a critical safety measure, ensuring that users can intervene quickly to prevent potential accidents or damage.

To meet the user's requirement for a 50-centimeter reach, modifications were made to the robot's design. The robot arm's length was extended to achieve the desired range, enabling it to cover a larger operational area. Additionally, a gripper was specifically designed and tailored to meet the robot's operational needs. This custom gripper ensures precise handling and placement of the golf balls, enhancing the robot's efficiency and reliability during operation.

By integrating these features—teach mode, sound feedback, an emergency stop function, and extended reach—the system achieves a balance of usability, safety, and functionality. These enhancements not only accommodate a wide range of user needs but also ensure a seamless and enjoyable interaction with the robot.

Reflection

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